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Gram's law and Selberg's conjecture on the distribution of zeros of the Riemann zeta function

M. A. Korolev

Abstract. We prove a number of new assertions related to the zeros of the Riemann zeta function $\zeta(s)$ and to the so-called Gram law.

Keywords: Riemann zeta function, Gram's law, Gram's rule, the argument of the Riemann zeta function.

Introduction

For a positive number t we define a function $\vartheta(t)$ as follows: $\vartheta(t)$ is equal to the increment in a continuous branch of the argument of the function $\pi^{-s/2}\Gamma(\frac{s}{2})$ as s varies along the segment connecting the points $\frac{1}{2}$ and $\frac{1}{2}+it$. By choosing the branch of the argument whose value at the point $s=\frac{1}{2}$ is equal to zero and using Stirling's formula for the gamma function, we obtain

$$\vartheta(t)=\operatorname{Im}\ln\Gamma\biggl(\frac{1}{4}+\frac{it}{2}\biggr)-\frac{t}{2}\ln\pi=\frac{t}{2}\ln\frac{t}{2\pi}-\frac{t}{2}-\frac{\pi}{8}+\Delta(t),$$

where, as $t\to+\infty$, the function $\Delta(t)$ expands into an asymptotic series of the form

$$\sum_{n=1}^{\infty}\frac{2^{2n-1}-1}{2^{2n}2n(2n-1)}(-1)^{n+1}B_{2n}t^{-(2n-1)},$$

where the B_{2n} are the Bernoulli numbers: $B_2=\frac{1}{6}$, $B_4=-\frac{1}{30}$, $B_6=\frac{1}{42}$, $B_8=-\frac{1}{30}$, $B_{10}=\frac{5}{66}$, and so on.

Now let $n\geqslant 0$ be an integer. We define the *Gram point* t_n to be the unique root of the equation

$$\vartheta(t_n)=\pi\cdot(n-1),$$

so that, for example,

$$t_0=9.666908\ldots,\qquad t_1=17.845599\ldots,\qquad t_2=23.170282\ldots,$$
$$t_3=27.670182\ldots,\qquad t_4=31.717979\ldots,\qquad t_5=35.467184\ldots.$$

For large n we have the asymptotic equations

$$t_n=\frac{2\pi n}{\ln n}(1+o(1)),\qquad t_{n+1}-t_n=\frac{2\pi}{\ln n}(1+o(1)).$$

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When calculating the first zeros of the Riemann zeta function $\zeta(s)$ at the beginning of the 20th century, Gram [1] discovered that for $1 \leqslant n \leqslant 15$ the inequalities

$$t_{n-1} < \gamma_n \leqslant t_n \tag{1}$$

hold, where the γ_n are the positive ordinates of the complex zeros of $\zeta(s)$ numbered in ascending order:

$$\gamma_1 = 14.134725\ldots, \quad \gamma_2 = 21.022039\ldots, \quad \gamma_3 = 25.010857\ldots,$$
$$\gamma_4 = 30.424876\ldots, \quad \gamma_5 = 32.935061\ldots, \quad \gamma_6 = 37.586178\ldots.$$

In other words, the ordinates of the first zeros of the Riemann zeta function are separated from one another by the Gram points.

However, Gram conjectured that this pattern is not general. Indeed, in 1925 Hutchinson [2] found the first two values of n for which the inequalities (1) do not hold, namely, $n = 127$ and $n = 136$, where

$$\gamma_{126} < t_{126} < t_{127} < \gamma_{127}, \qquad t_{134} < \gamma_{135} < \gamma_{136} < t_{135}.$$

Nevertheless, the inequalities (1) do hold in the vast majority of cases, and so Hutchinson called this pattern *Gram's rule*, or *Gram's law*.

The calculations of Gram and Hutchinson were continued by Titchmarsh (see [3], [4]), who established that for the first 1042 ordinates Gram's rule is violated in only 43 cases, less than 5% of the total number.

Such observations suggested to researchers that, although exceptions to Gram's rule do occur, they are nevertheless fairly rare, so that this pattern has the right to be called a 'law'.

Titchmarsh [3] was the first to pose the question of how 'strongly' the ordinates γ_n can 'deviate' from Gram's rule. By determining for a given $n \geqslant 1$ the number m from the inequality

$$t_{m-1} < \gamma_n \leqslant t_m,$$

Titchmarsh took as a measure of the deviation a certain quantity that is in essence close to the difference

$$\Delta_n = m - n,$$

and then proved that it is unbounded both above and below:

$$\varliminf_{n\to+\infty} \Delta_n = -\infty, \qquad \varlimsup_{n\to+\infty} \Delta_n = +\infty. \tag{2}$$

In other words, the 'deviation' thus defined can be arbitrarily large: the number n of the ordinate γ_n and the number m of the Gram interval $(t_{m-1}, t_m]$ containing this ordinate can differ by a number of arbitrarily large absolute value.

Nothing was known about how often exceptions to Gram's rule occur until the middle of the 1940s. The first results of this kind were obtained by Selberg, who stated them (without proof) in Part IV of his talk 'The zeta function and the Riemann hypothesis' [5] in 1946.

First, Selberg proved that for a sufficiently large N there exist at least KN Gram intervals $(t_{n-1}, t_n]$, $1 \leqslant n \leqslant N$, that do not contain any ordinates of the zeros of $\zeta(s)$, and that there are at least KN intervals $(t_{n-1}, t_n]$, $1 \leqslant n \leqslant N$, in each of which two or more ordinates occur (here K is some absolute constant).

Second, he established the following relations for any fixed integer $k \geqslant 1$:

$$\sum_{n=1}^{N} \Delta_n^{2k} = \frac{(2k)!}{k!} \frac{N}{(2\pi)^k} (\ln\ln N)^k + O\big(N(\ln\ln N)^{k-1/2}\big), \tag{3}$$

$$\sum_{n=1}^{N} \Delta_n^{2k-1} = O\big(N(\ln\ln N)^{k-1}\big). \tag{4}$$

Equation (3) shows that the average value of the quantity $|\Delta_n|$, $n = 1, 2, \ldots, N$, increases as $\sqrt{\ln\ln N}$, and the estimate (4) suggests that positive and negative values among the numbers Δ_n occur in approximately equal parts.

By setting $k = 1$ in (3) we find that

$$\sum_{n=1}^{N} \Delta_n^2 \sim \frac{N}{2\pi^2} \ln\ln N.$$

If in the sum over n we keep only the summands corresponding to the condition $|\Delta_n| > \Phi\sqrt{\ln\ln N}$, then, denoting by ν the number of terms in the new sum, we obtain

$$\nu\Phi^2 \ln\ln N \leqslant \sum_{n=1}^{N} \Delta_n^2 \sim \frac{N}{2\pi^2} \ln\ln N,$$

whence we have

$$\nu \ll N\Phi^{-2}.$$

By choosing the quantity Φ to depend on N in such a way that $\Phi(N) \to +\infty$ as N increases, we conclude that the number of those differences Δ_n whose absolute value is large compared with its average value $\sqrt{\ln\ln N}$ is of the form $O(N\Phi^{-2}) = o(N)$. In other words, for 'almost all' $n \leqslant N$ we have the inequality

$$|\Delta_n| \leqslant \Phi\sqrt{\ln\ln N}.$$

It is probable that considerations of this kind led Selberg to make the following conjecture (see [5], p. 199): *'It is probably true, though I have not been able to prove it rigorously, that $\sqrt{\log\log n}$ is the "normal" order of magnitude of Δ_n in the sense that if $\Phi(n)$ is a positive function of n which tends to infinity with n, the inequalities*

$$\frac{\sqrt{\log\log n}}{\Phi(n)} < |\Delta_n| < \Phi(n)\sqrt{\log\log n}$$

hold true for almost all n. In particular this should imply that γ_ν "almost never" lies in the interval $(t_{\nu-1}, t_\nu)$. It is the first inequality which is the difficult point, and which I have not been able to prove completely.'

In particular, Selberg's conjecture implies that the cases when Gram's rule still holds occur extremely rarely. The large proportion of such cases for small n is explained by the fact that in the domain covered by the initial calculations ($n \leqslant 1042$), the influence of the factor $\sqrt{\ln \ln n}$ is practically undetectable.

Our aim here is to prove the part of Selberg's conjecture which asserts that the ordinate γ_n 'almost never' occurs in the Gram interval $(t_{n-1}, t_n]$. We hope to devote a separate paper to the study of the distribution of the values of the quantities Δ_n.

We now briefly describe the ideas to be used in this paper. First, instead of Δ_n we introduce the quantities $\Delta(n)$ by the following rule: if for a given $n \geqslant 1$ the inequalities

$$\gamma_m \leqslant t_n < \gamma_{m+1}$$

hold, then we set $\Delta(n)$ equal to the difference $m - n$.

Generally speaking, the quantities Δ_n and $\Delta(n)$ are different: in the first case their numbering is connected with the ordinates γ_n, and in the second with the Gram points t_n. For example, in the case of the exceptions to Gram's rule found by Hutchinson we have $\Delta_{126} = 0$, $\Delta_{127} = 1$, while $\Delta(126) = 0$, $\Delta(127) = -1$, and $\Delta_{135} = 0$, $\Delta_{136} = -1$ but at the same time $\Delta(135) = 1$, $\Delta(136) = 1$. However, the violation of the inequalities $\gamma_n \leqslant t_n < \gamma_{n+1}$ for almost all n also implies the violation of inequalities (1) for almost all n; see Corollary 5. Thus, it is sufficient to show that the difference $\Delta(n)$ is non-zero for almost all n, and this problem reduces in turn to that of the behaviour of the argument of the Riemann zeta function at the Gram points.

This is carried out as follows. The Riemann–von Mangoldt formula expressing the number $N(T)$ of zeros of $\zeta(s)$ whose ordinate γ is positive and does not exceed T can be written in the form

$$N(T) = \pi^{-1}\vartheta(T) + 1 + S(T), \qquad S(T) = \pi^{-1} \arg \zeta\left(\frac{1}{2} + iT\right). \tag{5}$$

Here $\arg \zeta(\frac{1}{2} + iT)$ denotes the increment of any continuous branch of $\arg \zeta(s)$ along the polygonal arc connecting the points 2, $2 + iT$ and $\frac{1}{2} + iT$. The function $S(T)$ is called the *argument* of the Riemann zeta function on the critical line; its basic properties are reflected in the surveys [6], [7].

By taking $T = t_n$ in (5) and assuming that $\gamma_m \leqslant t_n < \gamma_{m+1}$ we obtain that

$$m = N(t_n) = \pi^{-1}\pi \cdot (n-1) + 1 + S(t_n) = n + S(t_n),$$

whence

$$\Delta(n) = m - n = S(t_n). \tag{6}$$

This relation is a crucial one for the present paper and it reduces the study of Gram's rule to the study of the values of the function $S(t)$ at the points t_n. For example, in view of (6), the modern omega-theorems for the argument of the Riemann zeta function on the critical line and the classical upper bounds for the quantity $|S(t)|$ imply that

$$\Delta(n) = \Omega_{\pm}\big((\ln n)^{1/3}(\ln \ln n)^{-1/3}\big), \qquad \Delta(n) = O(\ln n),$$

and if the Riemann hypothesis is true, then

$$\Delta(n) = \Omega_{\pm}\bigl((\ln n)^{1/2}(\ln\ln n)^{-1/2}\bigr), \qquad \Delta(n) = O\biggl(\frac{\ln n}{\ln\ln n}\biggr).$$

In particular, an analogue of the relations (2) in Titchmarsh's theorem holds for the quantities $\Delta(n)$:

$$\varliminf_{n\to+\infty} \Delta(n) = -\infty, \qquad \varlimsup_{n\to+\infty} \Delta(n) = +\infty.$$

This, however, does not exhaust the analogy between the quantities $\Delta(n)$ and Δ_n. It turns out that the numbers $\Delta(n)$ satisfy relations similar to formulae (3), (4) in Selberg's conjecture.

It is natural to ask whether an analogue of Selberg's conjecture holds for the quantities $\Delta(n)$. The answer turns out to be affirmative. Whatever positive, monotonic, and unboundedly increasing function we take for $\Phi(x)$, for 'almost all' n we have the inequalities

$$\frac{\sqrt{\ln\ln n}}{\Phi(n)} < |\Delta(n)| < \Phi(n)\sqrt{\ln\ln n}. \tag{7}$$

In particular, $\Delta(n)$ 'almost never' vanishes, that is, the point t_n 'almost never' occurs in the interval $[\gamma_n, \gamma_{n+1})$. As pointed out above, this also implies Selberg's conjecture that $\Delta_n \neq 0$ for almost all n.

Finally we clarify the technical aspect of this paper. To do this we consider the discrete random quantity δ taking the values

$$\delta(n) = \frac{\Delta(n)}{\sqrt{\frac{1}{2\pi^2}\ln\ln N}}, \qquad N < n \leqslant N+M, \quad 0 < M < \frac{1}{2}N.$$

The distribution function $F(x)$ of this random quantity expresses the proportion of the numbers in the sequence $N+1,\dots,N+M$ corresponding to the condition $\delta(n) \leqslant x$ or, which is the same, the condition $\Delta(n) \leqslant x\sqrt{\frac{1}{2\pi^2}\ln\ln N}$. Given a positive function $\Phi(n)$ with the above properties, it is easy to verify that the number m of integers n satisfying either of the inequalities

$$|\Delta(n)| \geqslant \Phi(n)\sqrt{\ln\ln n}, \qquad |\Delta(n)| \leqslant \frac{1}{\Phi(n)}\sqrt{\ln\ln n}$$

does not exceed the total number of solutions of the inequalities

$$|\delta(n)| > \pi\sqrt{2}\,\Phi(N), \qquad |\delta(n)| < \frac{2\pi\sqrt{2}}{\Phi(N)},$$

which in turn is equal to the sum $m_1 + m_2$, where

$$m_1 = M\bigl(1 - F(\pi\sqrt{2}\Phi)\bigr) + MF(-\pi\sqrt{2}\Phi),$$

$$m_2 = M\biggl(F\biggl(\frac{2\pi\sqrt{2}}{\Phi}\biggr) - F\biggl(-\frac{2\pi\sqrt{2}}{\Phi}\biggr)\biggr), \qquad \Phi = \Phi(N).$$

One of the assertions in this paper is that the distribution $F(x)$ proves to be very close to the Gaussian distribution

$$G(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-u^2/2} du.$$

This fact and the easily verifiable estimates

$$1 - G(\pi\sqrt{2}\Phi) + G(-\pi\sqrt{2}\Phi) \leqslant \frac{e^{-\pi^2\Phi^2}}{\pi\sqrt{\pi}\Phi}, \qquad G\left(\frac{2\pi\sqrt{2}}{\Phi}\right) - G\left(-\frac{2\pi\sqrt{2}}{\Phi}\right) < \frac{4\sqrt{\pi}}{\Phi}$$

will imply that

$$m_1 = o(M), \qquad m_2 = o(M), \qquad m_1 + m_2 = o(M)$$

as $N \to +\infty$, that is, the inequalities (7) are true for almost all n.

An approximate formula for the distribution function $F(x)$ is obtained using the following method, which was first applied in the study of the argument of the zeta function by Ghosh [8]. Let $f(t)$ be the characteristic function of the desired distribution, that is,

$$f(t) = \int_{-\infty}^{+\infty} e^{itu}\, dF(u) = M^{-1} \sum_{N<n\leqslant N+M} e^{it\delta(n)}.$$

If we expand each of the exponentials $e^{it\delta(n)}$ into a Taylor series and make the summation over n inner, then $f(t)$ is represented (with some error) by the finite sum

$$M^{-1} \sum_{N<n\leqslant N+M} \sum_{k} \frac{(it)^k}{k!} \delta^k(n) = \sum_{k} \frac{(it)^k}{k!} M^{-1} \sum_{N<n\leqslant N+M} \delta^k(n).$$

Based on the asymptotic formulae for the quantities

$$j_k(N, M) = M^{-1} \sum_{N<n\leqslant N+M} \delta^k(n),$$

we shall show that the sum representing $f(t)$ is very close to the Taylor series of the function $g(t) = e^{-t^2/2}$, the characteristic function of the distribution $G(x)$. Given an estimate for the difference $|f(t)-g(t)|$, we can use the Berry–Esseen inequality to estimate the deviation of the desired function $F(x)$ from the Gaussian distribution.

Thus in the end, the proof of Selberg's conjecture reduces to calculating the sums $j_k(N, M)$ or, which is the same, sums of the form

$$J_k(N, M) = \sum_{N<n\leqslant N+M} \Delta_n^k = \sum_{N<n\leqslant N+M} (S(t_n))^k, \qquad k = 1, 2, 3, \ldots .$$

Problems related to summing the values of the functions $S(t)$ and $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ at the Gram points and at the ordinates of the zeros of $\zeta(s)$ constitute the subject of the discrete theory of the Riemann zeta function. In particular, the

papers [9]–[16] are devoted to an investigation of these problems (as well as their generalizations to the case of Dirichlet L-functions). Here we mention only one of Fujii's results: as N increases unboundedly and an integer $k > 1$ increases with N, the distribution function of the discrete random quantity with the values

$$\frac{\pi(S(t_{n+k}) - S(t_n))}{\sqrt{\ln k}} = \frac{\pi(N(t_{n+k}) - N(t_n) - k)}{\sqrt{\ln k}}, \qquad n = 1, 2, \dots, N,$$

has the Gaussian distribution $G(x)$ as its limit.

The quantities $J_k(N, M)$, as well as the sums

$$I_k(N, M) = \sum_{N < n \leqslant N+M} |\Delta(n)|^k, \qquad k > 0,$$

are discrete analogues of the integrals

$$\int_T^{T+H} (S(t))^k \, dt, \qquad \int_T^{T+H} |S(t)|^k \, dt, \tag{8}$$

which are called the *moments of the function* $S(t)$. Therefore asymptotic expressions for such sums can be obtained by the same method as used for the asymptotics of the corresponding integrals (see [17], [18]). They are derived on the basis of the so-called 'explicit formula' of Selberg, which makes it possible in the course of the calculations to replace (with admissible accuracy) the quantity $S(t)$ by the sum

$$\frac{1}{\pi} \sum_{p<y} \frac{\sin(t \ln p)}{\sqrt{p}} = \frac{1}{2\pi i} \left(\sum_{p<y} \frac{p^{it}}{\sqrt{p}} - \sum_{q<y} \frac{q^{-it}}{\sqrt{q}} \right),$$

extended to all the primes in $(1, y]$, $y = y(t)$.

The calculation of the sums $I_k(N, M)$, $J_k(N, M)$ and their integral analogues (8) is based on the so-called *density theorem* for the zeros of the Riemann zeta function, that is, on a non-trivial estimate for the difference

$$N(\sigma, T + H) - N(\sigma, T - H), \qquad \frac{1}{2} \leqslant \sigma \leqslant 1,$$

which expresses the number of zeros of $\zeta(s)$ in the rectangle

$$\sigma < \operatorname{Re} s \leqslant 1, \qquad T - H < \operatorname{Im} s \leqslant T + H$$

(see, for example, [17], [19], [20]). Here both the smallest value H of the length of the interval of integration in (8) and the minimal number M of terms in the sums $I_k(N, M)$, $J_k(N, M)$ (that is, the most interesting cases) are determined by the 'capabilities' of the density theorem and are of the same order:

$$M = [N^{27/82+\varepsilon}], \qquad H = T^{27/82+\varepsilon}$$

for all $N \geqslant N_0(\varepsilon)$, $T \geqslant T_0(\varepsilon)$, and

$$M = [N^{\varepsilon}], \qquad H = T^{\varepsilon}$$

for 'almost all' $N \geqslant N_1(\varepsilon)$, $T \geqslant T_1(\varepsilon)$, where $[\,\cdot\,]$ is the integer part of a number.

The paper consists of five sections. In § 1 we give the statements and proofs of the auxiliary assertions needed in the proofs of the main lemmas and theorems. In § 2 we obtain upper bounds for the discrete analogues of integrals of a special form, that is, for the sums

$$\sum_{N<n\leqslant N+M}\left|\sum_{p\leqslant y}\frac{a(p)}{\sqrt{p}}p^{it_n}\right|^{2m}$$

and the like. Here we make substantial use of the results obtained in §§ 2, 3 of Ch. II in [7] in a number of cases omitting some stages in the calculations. In §§ 3 and 4 we derive the asymptotic formulae and upper bounds for the sums $I_k(N,M)$ and $J_k(N,M)$ in the case of integer and fractional values of k. Finally, § 5 is devoted to proving Selberg's conjecture.

Throughout the paper, ε denotes an arbitrarily small fixed positive number not exceeding 10^{-3}; $\theta,\theta_1,\theta_2,\dots$ are complex numbers whose absolute values do not exceed 1 and which are, generally speaking, different in different relations; $\Lambda(n)$ is the von Mangoldt function, which is equal to $\ln p$ when $n=p^k$, where $k=1,2,\dots$ and p is a prime, and to zero otherwise; $\operatorname{mes}E$ denotes the Lebesgue measure of a set E and $\{x\}$ the fractional part of a number x.

§ 1. Auxiliary lemmas

Lemma 1. *Let $l\geqslant 2$ be an integer. Then the following inequalities hold*:

$$\sum_{1\leqslant k<l}\frac{1}{k}\frac{1}{\ln\frac{l}{k}}<4\ln l,\qquad \sum_{1\leqslant k<l}\frac{1}{\sqrt{k}}\frac{1}{\ln\frac{l}{k}}<4\sqrt{l}\ln l.$$

Proof. By using the obvious inequality

$$\ln\frac{l}{k}=-\ln\left(1-\frac{l-k}{l}\right)>\frac{l-k}{l}$$

for $l>k$, we find that

$$\sum_{1\leqslant k<l}\frac{1}{k}\frac{1}{\ln\frac{l}{k}}<\sum_{1\leqslant k<l}\frac{l}{k(l-k)}=2\sum_{1\leqslant k<l}\frac{1}{k}$$
$$<2\sum_{1\leqslant k<l}\bigl(\ln(2k+1)-\ln(2k-1)\bigr)=2\ln(2l-1)<4\ln l.$$

For the second sum, we obtain

$$\sum_{1\leqslant k<l}\frac{1}{\sqrt{k}}\frac{1}{\ln\frac{l}{k}}=\sum_{1\leqslant k<l}\frac{\sqrt{k}}{k}\frac{1}{\ln\frac{l}{k}}<\sqrt{l}\sum_{1\leqslant k<l}\frac{1}{k}\frac{1}{\ln\frac{l}{k}}<4\sqrt{l}\ln l.$$

The lemma is proved.

Lemma 2. *Let $V(t)$ be a complex-valued function that is continuously differentiable on the segment $[\tau_0, \tau_k]$, and let $\tau_0 < \tau_1 < \cdots < \tau_{k-1} < \tau_k$. Then for*

$$\delta = \min_{0 \leqslant r < k} (\tau_{r+1} - \tau_r),$$

we have

$$\sum_{r=1}^{k} |V(\tau_r)|^2 \leqslant \frac{1}{\delta} \int_{\tau_0}^{\tau_k} |V(t)|^2 \, dt + 2 \left(\int_{\tau_0}^{\tau_k} |V(t)|^2 \, dt \right)^{1/2} \left(\int_{\tau_0}^{\tau_k} |V'(t)|^2 \, dt \right)^{1/2}.$$

For a proof, see, for example, [21], Russian pp. 122, 123.

Lemma 3 (van der Corput). *Let $f(x)$ be a real differentiable function in an interval $M < x < M'$ such that inside the interval its derivative $f'(x)$ is monotonic, sign-constant, and satisfies the inequality $|f'(x)| \leqslant \delta$ for a constant δ such that $0 < \delta < 1$. Then we have*

$$\sum_{M < x \leqslant M'} e^{2\pi i f(x)} = \int_{M}^{M'} e^{2\pi i f(x)} \, dx + \theta \left(3 + \frac{2\delta}{1 - \delta} \right).$$

For a proof, see [22], Russian pp. 25–27.

Let $n \geqslant 2$ be an integer, let $z \geqslant 2$, and let

$$\Lambda_z(n) = \begin{cases} \Lambda(n) & \text{if } 1 \leqslant n \leqslant z, \\ \Lambda(n) \dfrac{\ln^2(z^3/n) - 2\ln^2(z^2/n)}{2\ln^2 z} & \text{if } z < n \leqslant z^2, \\ \Lambda(n) \dfrac{\ln^2(z^3/n)}{2\ln^2 z} & \text{if } z^2 < n \leqslant z^3. \end{cases}$$

Next let $2 \leqslant z \leqslant t^2$. Following the notation introduced by Selberg, we set

$$\sigma_{z,t} = 0.5 + 2 \max(|\beta - 0.5|, \ln^{-1} z),$$

where the maximum is taken over all the zeros $\varrho = \beta + i\gamma$ of the Riemann zeta function satisfying the condition

$$|t - \gamma| \leqslant \frac{z^{3|\beta - 0.5|}}{\ln z}.$$

Lemma 4 (Selberg's 'explicit formula'). *Let $e^{16} < z \leqslant t^2$. Then*

$$S(t) = -\frac{1}{\pi} \sum_{n \leqslant z^3} \frac{\Lambda_z(n)}{n^{\sigma_{z,t}}} \frac{\sin(t \ln n)}{\ln n} + 15\theta(\sigma_{z,t} - 0.5)(|r(z,t)| + \ln|t|),$$

where

$$r(z,t) = \sum_{n \leqslant z^3} \frac{\Lambda_z(n)}{n^{\sigma_{z,t} + it}}.$$

For a proof, see [17].

Lemma 5. *Let $m \geqslant 1$, $y > e^3$, $\tau > 0$, and let $p_1, \dots, p_m$, $q_1, \dots, q_m$ take values of prime numbers in the interval $(1, y]$ and satisfy the condition $p_1 \cdots p_m = q_1 \cdots q_m$. Then for any sequence $a(n)$ of complex numbers, the sum*

$$\Sigma = \sum_{\substack{p_1,\dots,p_m \\ q_1,\dots,q_m}} \frac{a(p_1)\cdots a(p_m)\overline{a}(q_1)\cdots\overline{a}(q_m)}{(p_1\cdots p_m q_1\cdots p_m)^{\tau}}$$

satisfies the formula

$$\Sigma = m!\,\Sigma_1^m + \theta m^2 m!\,\Sigma_1^{m-2}\Sigma_2,$$

where

$$\Sigma_j = \sum_{p\leqslant y}\left(\frac{|a(p)|}{p^{\tau}}\right)^{2j}, \qquad j = 1, 2,$$

and $|\theta| \leqslant 1$ for $m > 1$, and $\theta = 0$ for $m = 1$.

For a proof, see [7], [18].

§ 2. Basic assertions

Lemma 6. *Suppose that $0 < \kappa < \frac{1}{2}$, $0 < c < \frac{1}{2} - \kappa$, k, r, ν, N, M are integers such that $k \geqslant 0$, $r \geqslant 0$, $k + r = \nu$, $\nu \geqslant 1$, $N \geqslant \exp\{9\kappa^{-1}\}$, $M \geqslant \exp\{1.5\nu c^{-1}\}$ and $M \leqslant N$, and let $y = M^{2c/\nu}$. Furthermore, let $p_1, \dots, p_k$, $q_1, \dots, q_r$ take values of prime numbers in the interval $(1, y]$ and satisfy the condition $p_1 \cdots p_k \neq q_1 \cdots q_r$. Finally, suppose that all the terms of a sequence $a(p)$ for $p \leqslant y$ satisfy the condition $|a(p)| \leqslant \delta$. If t_n is the Gram sequence, then the sum*

$$S = \sum_{N<n\leqslant N+M} \sum_{\substack{p_1,\dots,p_k \\ q_1,\dots,q_r}} \frac{a(p_1)\cdots a(p_k)\overline{a}(q_1)\cdots\overline{a}(q_r)}{(p_1\cdots q_r)^{1/2}} \left(\frac{q_1\cdots q_r}{p_1\cdots p_k}\right)^{it_n}$$

satisfies the bound $|S| < (\delta y^{3/2})^{\nu}\ln N$.

Proof. By making the summation over n inner and letting S_1 denote the maximum of the absolute values of the inner sums taken over all the admissible tuples $p_1, \dots, p_k$, $q_1, \dots, q_r$, we obtain

$$|S| \leqslant \delta^{\nu} S_1 \sum_{\substack{p_1,\dots,p_k \\ q_1,\dots,q_r}} (p_1\cdots q_r)^{-1/2} \leqslant \delta^{\nu} S_1 \left(\sum_{p\leqslant y}\frac{1}{\sqrt{p}}\right)^{\nu} < \delta^{\nu} S_1 \left(\frac{3\sqrt{y}}{\ln y}\right)^{\nu} < (\delta\sqrt{y})^{\nu} S_1.$$

Let $a > 1$ be that of the fractions $\frac{q_1\cdots q_r}{p_1\cdots p_k}$ at which the largest value of S_1 is attained. Then

$$S_1 = \left|\sum_{N<n\leqslant N+M} e^{2\pi i f(n)}\right|, \qquad f(x) = \frac{\ln a}{2\pi}\, t_x,$$

where t_x denotes the unique solution of the equation $\vartheta(t_x) = \pi\cdot(x-1)$ for $x \geqslant 1$. By differentiating this equation with respect to x and observing that

$$t_x = \frac{2\pi x}{\ln x}(1 + o(1))$$

for $N \leqslant x \leqslant N+M$, we find that

$$t'_x = \frac{\pi}{\vartheta'(t_x)} = \frac{\pi}{\frac{1}{2}\ln\frac{t_x}{2\pi} + \Delta'(t_x)} = \frac{2\pi}{\ln x}(1+o(1)),$$

$$|f'(x)| = \frac{\ln a}{2\pi}t'_x = \frac{\ln a}{\ln x}(1+o(1)) \leqslant \frac{\ln y^\nu}{\ln x}(1+o(1)) \leqslant 2c\frac{\ln M}{\ln N}(1+o(1))$$
$$\leqslant (1-2\kappa)(1+o(1)) < 1-\kappa.$$

By Lemma 3 this implies the inequality

$$S_1 \leqslant |j| + 3 + 2\kappa^{-1}(1-\kappa) \leqslant |j| + 3\kappa^{-1},$$

where

$$j = \int_N^{N+M} e^{2\pi i f(x)}\,dx.$$

Integrating by parts and using the monotonicity of the function $1/f'(x)$, we now obtain

$$|j| = \frac{1}{2\pi}\left|\frac{e^{2\pi i f(N+M)}}{f'(N+M)} - \frac{e^{2\pi i f(N)}}{f'(N)} - \int_N^{N+M} e^{2\pi i f(x)}\,d\frac{1}{f'(x)}\right|$$
$$\leqslant \frac{1}{2\pi}\left(\frac{1}{f'(N+M)} + \frac{1}{f'(N)} + \frac{1}{f'(N+M)} - \frac{1}{f'(N)}\right) = \frac{1}{\pi f'(N+M)}$$
$$= \frac{\ln(N+M)}{\pi\ln a}(1+o(1)) \leqslant \frac{\ln N}{3\ln a}.$$

Finally, by observing that

$$\ln a \geqslant \ln\frac{p_1\cdots p_k+1}{p_1\cdots p_k} \geqslant \frac{1}{2}(p_1\cdots p_k)^{-1} \geqslant \frac{1}{2}y^{-k} \geqslant \frac{1}{2}y^{-\nu},$$

we obtain the inequalities

$$|j| \leqslant \frac{2}{3}y^\nu\ln N, \qquad S_1 < \frac{2}{3}y^\nu\ln N + 3\kappa^{-1} < y^\nu\ln N, \qquad |S| < (\delta y^{3/2})^\nu\ln N.$$

The lemma is proved.

Lemma 7. *Suppose that $B>0$, $0<c\leqslant\frac{1}{2}$, M, N are integers such that $N\geqslant N_0(\varepsilon)$, $N^\varepsilon \leqslant M \leqslant \sqrt[3]{N}$, $M \geqslant \exp\{e^2c^{-1}\}$, m is an integer such that $1\leqslant m\leqslant ce^{-2}\ln M$, and let $y = M^{c/m}$. Furthermore, let p take values of prime numbers in the interval $(1,y]$, and suppose that a sequence $a_1(p)$ for $p\leqslant y$ satisfies the condition $|a_1(p)|\leqslant B\frac{\ln p}{\ln y}$. Then the sum*

$$S_1 = \sum_{N<n\leqslant N+M}\left|\sum_{p\leqslant y}\frac{a_1(p)}{\sqrt{p}}p^{it_n}\right|^{2m}$$

satisfies the bound $S_1 \leqslant 3.2(2B^2m)^mM$.

Proof. We define $b(k)$ as the sum of the quantities $a_1(p_1)\cdots a_1(p_m)$ extended to all the representations of an integer k in the form of a product $p_1\cdots p_m$ of prime numbers not exceeding y. Then the original sum takes the form

$$\sum_{N<n\leqslant N+M}|V(t_n)|^2,\quad \text{where}\quad V(t)=\sum_{1\leqslant k\leqslant Y}\frac{b(k)}{\sqrt{k}}k^{it},\quad Y=y^m=M^c.$$

It follows from Lemma 2 that the upper bound for S_1 does not exceed the quantity

$$\delta^{-1}j_1+2\sqrt{j_1}\,\sqrt{j_2}\,,$$

in which j_1 and j_2 denote the integrals of the functions $|V(t)|^2$ and $|V'(t)|^2$ over the segment $T\leqslant t\leqslant T+H$, where $T=t_N$, $H=t_{N+M}-t_N$, and δ is taken to be equal to the least of the differences $t_{r+1}-t_r$, where $r=N,\,N+1,\dots,N+M-1$, that is, $t_{N+M}-t_{N+M-1}$. By transforming the first of these integrals we obtain

$$j_1=H\Sigma_1-2\operatorname{Im}\Sigma_2,$$

$$\Sigma_1=\sum_{k\leqslant Y}\frac{|b(k)|^2}{k},\qquad \Sigma_2=\sum_{1<l\leqslant Y}\sum_{1\leqslant k\leqslant l-1}\frac{b(k)\bar{b}(l)}{\sqrt{kl}}\,\frac{1}{\ln\frac{l}{k}}\big((kl^{-1})^{i(T+H)}-(kl^{-1})^{iT}\big).$$

It follows from the definition of $b(k)$ that

$$|b(k)|\leqslant\sum_{p_1\cdots p_m=k}\frac{B\ln p_1}{\ln y}\cdots\frac{B\ln p_m}{\ln y}\leqslant B^m m!\leqslant(Bm)^m.$$

By taking this estimate into account and using Lemma 1 we find that

$$\begin{aligned}\Sigma_1&\leqslant(Bm)^m\sum_{k\leqslant Y}\frac{|b(k)|}{k}\leqslant(Bm)^m\Bigg(\sum_{p\leqslant y}\frac{|a_1(p)|}{p}\Bigg)^m\\&\leqslant(Bm)^m\Bigg(\frac{B}{\ln y}\sum_{p\leqslant y}\frac{\ln p}{p}\Bigg)^m\leqslant(2B^2m)^m,\end{aligned}$$

$$\begin{aligned}|\Sigma_2|&\leqslant 2(Bm)^m\sum_{1<l\leqslant Y}\frac{|b(l)|}{\sqrt{l}}\sum_{1\leqslant k\leqslant l-1}\frac{1}{\sqrt{k}}\,\frac{1}{\ln\frac{l}{k}}<2(Bm)^m\sum_{1<l\leqslant Y}\frac{|b(l)|}{\sqrt{l}}4\sqrt{l}\ln l\\&\leqslant 8(Bm)^m m\ln y\Bigg(\sum_{p\leqslant y}|a_1(p)|\Bigg)^m\leqslant 8m(Bm)^m\ln y\Bigg(\frac{B}{\ln y}\sum_{p\leqslant y}\ln p\Bigg)^m\\&\leqslant 8(B^2m)^m\frac{m}{(\ln y)^{m-1}}(2y)^m.\end{aligned}$$

The hypotheses of the lemma imply the inequalities

$$\ln y\geqslant\frac{c\ln M}{ce^{-2}\ln M}=e^2,$$

$$Y=y^m=M^c\leqslant\sqrt{M}<\sqrt{\frac{1}{6}H\ln N}\leqslant\sqrt{\frac{1}{6}\cdot 2\varepsilon^{-1}H\ln H}<\sqrt{H}\ln H,$$

which lead to the estimate

$$|\Sigma_2| \leqslant 8(2B^2m)\frac{m}{e^{2(m-1)}}\sqrt{H}\ln H \leqslant 8(2B^2m)^m\sqrt{H}\ln H.$$

Thus, for a sufficiently small $\varepsilon_1 > 0$ we can write an upper bound for j_1 in the form

$$j_1 \leqslant (2B^2m)^m H + 16(2B^2m)^m\sqrt{H}\ln H < U, \tag{9}$$

where $U = (1+\varepsilon_1)(2B^2m)^m H$.

The integral j_2 can be represented in the form $H\Sigma_1' - 2\operatorname{Im}\Sigma_2'$, where the sums Σ_1' and Σ_2' are obtained from the expressions for Σ_1 and Σ_2 by replacing the factors $b(k)$ and $b(l)$ by $b(k)\ln k$ and $b(l)\ln l$, respectively. Consequently, the upper bound for the integral j_2 differs from the estimate (9) by the factor $(\ln Y)^2 = (m\ln y)^2 = (c\ln M)^2$:

$$j_2 < U(\ln Y)^2 \leqslant U(c\ln M)^2 \leqslant U\left(\frac{1}{2}\ln M\right)^2 \leqslant U\left(\frac{1}{6}\ln N\right)^2.$$

We now return to estimating the sum S_1. By observing that

$$\delta = t_{M+N} - t_{M+N-1} > \frac{2\pi}{\ln N}(1+\varepsilon_1)^{-1}$$

as N increases, we find that

$$S_1 < \left((1+\varepsilon_1)(2\pi)^{-1}\ln N + 2\cdot\frac{1}{6}\ln N\right)U < \frac{1}{2}(2B^2m)^m H\ln N < \frac{16}{5}(2B^2m)^m M.$$

The lemma is proved.

Lemma 8. *Suppose that under the same restrictions on c, M, N, m, y, p as in Lemma 7, a sequence $a_2(p)$ satisfies the condition $|a_2(p)| \leqslant B$. Then the sum*

$$S_2 = \sum_{N<n\leqslant N+M}\left|\sum_{p\leqslant y}\frac{a_2(p)}{p}\,p^{2it_n}\right|^{2m}$$

satisfies the inequality $S_2 \leqslant (2B^2m)^m M$.

Proof. Defining numbers $c(k)$ by analogy with $b(k)$ and setting $Y = y^m$ as above, we find that

$$|c(k)| \leqslant B^m m! \leqslant (Bm)^m, \qquad S_2 \leqslant \delta^{-1}j_1 + 2\sqrt{j_1}\,\sqrt{j_2}, \qquad j_1 = H\Sigma_1 - \operatorname{Im}\Sigma_2,$$

$$\Sigma_1 = \sum_{k\leqslant Y}\frac{|c(k)|^2}{k^2} \leqslant (Bm)^m\left(\sum_{p\leqslant y}\frac{|a_2(p)|}{p^2}\right)^m < (0.6B^2m)^m,$$

$$|\Sigma_2| \leqslant 2(Bm)^m\sum_{1<l\leqslant Y}\frac{|c(l)|}{l}\sum_{1\leqslant k<l}\frac{1}{k\ln\frac{l}{k}} < 8(Bm)^m\sum_{1<l\leqslant Y}\frac{|c(l)|}{l}\ln l$$

$$\leqslant 8(Bm)^m(c\ln M)\left(\sum_{p\leqslant y}\frac{|a_2(p)|}{p}\right)^m \leqslant 4(2B^2m)^m(\ln M)(\ln\ln y)^m.$$

We define a quantity λ by the equation $m = \frac{c}{e^2\lambda}\ln M$. It follows from the hypotheses of the lemma that, first, $\lambda \geqslant 1$ and, second, $\ln y = cm^{-1}\ln M = e^2\lambda$. Furthermore, using the fact that the maximum of the function $x^{-1}\ln\ln x$ for $x \geqslant e^2$ does not exceed $\frac{1}{10}$ we conclude that

$$(\ln\ln y)^m = \exp\left\{\frac{c}{e^2\lambda}\ln M\ln\ln(e^2\lambda)\right\} \leqslant \exp\left\{\frac{c}{10}\ln M\right\} \leqslant M^{0.1}$$

and, consequently, $|\Sigma_2| < 2(2B^2m)^m M^{0.1}\ln M$. Thus, for a sufficiently small $\varepsilon_1 > 0$ the upper bounds for the quantities j_1, j_2, S_2 can be written in the form

$$\begin{aligned} j_1 &< (0.6B^2m)^m H + 4(2B^2m)^m M^{0.1}\ln M \\ &\leqslant (2B^2m)^m((0.3)^m H + 4M^{0.1}\ln M) \leqslant (0.3+\varepsilon_1)(2B^2m)^m H, \\ j_2 &< (0.3+\varepsilon_1)(2B^2m)^m H(c\ln N)^2, \\ S_2 &\leqslant (\delta^{-1} + 2c\ln M)(0.3+\varepsilon_1)(2B^2m)^m H \\ &< \left((1+\varepsilon_1)(2\pi)^{-1}\ln N + \frac{1}{3}\ln N\right)(0.3+\varepsilon_1)(2B^2m)^m H \\ &< 0.16(2B^2m)^m H\ln N < (2B^2m)^m M. \end{aligned}$$

The lemma is proved.

In problems related to $S(t)$ the main role is played by density theorems of the form

$$N(\sigma, T+H) - N(\sigma, T-H) \leqslant cH(\ln T)x^{-(2\sigma-1)}, \qquad \frac{1}{2} \leqslant \sigma \leqslant 1, \tag{10}$$

where σ is an arbitrary number with the condition $\frac{1}{2} \leqslant \sigma \leqslant 1$, $H = T^\alpha$, $x = T^\beta$, and α, β, c are some constants satisfying $0 < \beta < \alpha \leqslant 1$ and $c > 0$. We give several variants of the estimate (10) that will be needed in what follows.

Karatsuba's Density Theorem [7], [19]. *Inequality* (10) *holds for any* σ *with the condition* $\frac{1}{2} \leqslant \sigma \leqslant 1$ *for all* $T \geqslant T_0(\varepsilon) > 0$ *with* $H = T^{27/82+\varepsilon}$, $x = T^{0.1\varepsilon}$ *and* $c = 13$.

Corollary 1. *Let* N *be an integer and let* $M = [N^{27/82+\varepsilon}]$. *Then inequality* (10) *holds for all* $N \geqslant N_0(\varepsilon) > 1$ *with* $T = t_N$, $H = t_{N+M} - t_N$, $x = T^{0.1\varepsilon}$ *and* $c = 13$.

Density Theorem ('for almost all T' [7], [20]). *Let* $X \geqslant X_0(\varepsilon) > 0$, $H = X^\varepsilon$, $x = H^{0.05}$ *and* $c = 13$. *Then inequality* (10) *holds for any* σ *with the condition* $\frac{1}{2} \leqslant \sigma \leqslant 1$ *for all* T *in* $(X, 2X]$, *except for a set* $\mathcal{E} = \mathcal{E}(X)$ *of values with measure* $\operatorname{mes}\mathcal{E} = O(X^{1-0.01\varepsilon})$.

Corollary 2. *Let* K *be an integer,* $K \geqslant K_0(\varepsilon) > 1$, *and let* $M = [K^\varepsilon]$, $x = M^{0.05}$, $c = 13$. *Then inequality* (10) *holds for any* σ *with the condition* $\frac{1}{2} \leqslant \sigma \leqslant 1$ *for all* N, $K < N \leqslant 2K$, $T = t_N$, $H = t_{N+M} - t_N$, *except for* $O(K^{1-0.01\varepsilon})$ *values.*

Throughout what follows, in this and subsequent sections, we assume that the numbers N, M, T, and H satisfy the conditions of one of the two density theorems above (or of their corollaries). Thus, all the assertions proved below hold

1) for all $N \geqslant N_0(\varepsilon) > 1$ with $M = [N^{27/82+\varepsilon}]$;

2) for all N in an arbitrary interval $K < N \leqslant 2K$, $K \geqslant K_0(\varepsilon) > 1$, except for $O(K^{1-0.01\varepsilon})$ values, with $M = [K^{\varepsilon}]$.

Throughout what follows in the statements and proofs of our assertions, for brevity we agree to use the notation $\sigma_z(n)$ instead of σ_{z,t_n}. We also point out that it is in the proof of the following lemma that the density theorem for the zeros of the Riemann zeta function is used.

Lemma 9. *Suppose that m, ν are integers, $m \geqslant 1$, $0 \leqslant \nu \leqslant 8m$, $z \geqslant 2$, $1 \leqslant \xi \leqslant z^{4m}$, $z^3\xi^2 \leqslant x$, and let*

$$V = \sum_{N<n\leqslant N+M} (\sigma_z(n) - 0.5)^{\nu} \xi^{\sigma_z(n)-0.5}.$$

Then we have the estimate

$$V < M\frac{(2e)^{8m}}{(\ln z)^{\nu}}\left(1 + \frac{\nu!}{18}\left(\frac{\ln z}{\ln x}\right)^{\nu}\frac{\ln N}{\ln z}\right).$$

Proof. We partition all the numbers n satisfying $N < n \leqslant N + M$ into two classes by including in the first class those for which any zero $\beta + i\gamma$ with the condition

$$|t_n - \gamma| \leqslant \frac{z^{3|\beta-0.5|}}{\ln z}$$

also satisfies the inequality $|\beta - 0.5| \leqslant \ln^{-1} z$; all the other numbers n are included in the second class. Then the sum V_1 over all the n in the first class does not exceed the quantity

$$\left(\frac{2}{\ln z}\right)^{\nu} \xi^{2/\ln z} M \leqslant \frac{(2e)^{8m}M}{(\ln z)^{\nu}}.$$

Next, for any point t_n corresponding to a value of n in the second class there exists a zero $\beta_0 + i\gamma_0$ with the condition

$$\gamma_0 - \delta_0 \leqslant t_n \leqslant \gamma_0 + \delta_0, \qquad \delta_0 = \frac{z^{3|\beta_0-0.5|}}{\ln z},$$

so that $|\beta_0 - 0.5| > \ln^{-1} z$. Consequently, all such Gram points t_n will be covered if each of the ordinates γ in the interval $(T - \sqrt{x}, T + H + \sqrt{x}\,]$, $T = t_N$, $H = t_{M+N} - t_N$, is surrounded by the neighbourhood $(\gamma - \delta, \gamma + \delta)$, where $\delta = z^{3|\beta-0.5|}\ln^{-1} z$ (for any of the zeros $\beta + i\gamma$ under consideration we obviously have $\delta \leqslant z^{1.5}\ln^{-1} z \leqslant \sqrt{x}$). Estimating the sum V_2 taken over all the n in the second class, we obtain

$$V_2 \leqslant 2 \sum_{\substack{T-\sqrt{x}<\gamma\leqslant T+H+\sqrt{x} \\ \beta>0.5}} \sum_{\gamma-\delta\leqslant t_n\leqslant\gamma+\delta} 2^{\nu}(\beta - 0.5)^{\nu}\xi^{2(\beta-0.5)}.$$

Let $t_{k+1}, t_{k+2}, \dots, t_{k+r}$ be all the Gram points that occur on the segment $[\gamma - \delta, \gamma + \delta]$. By Lagrange's finite increment formula, for some η, $\gamma - \delta < \eta < \gamma + \delta$, we have

$$r = \pi^{-1}\bigl(\vartheta(t_{k+r}) - \vartheta(t_{k+1})\bigr) + 1 = \pi^{-1}(t_{k+r} - t_{k+1})\vartheta'(\eta) + 1$$
$$\leqslant 2\pi^{-1}\delta\left(\frac{1}{2}\ln\frac{t_{N+M}}{2\pi} + O(t_{N+M}^{-1})\right) + 1 < \frac{1}{3}\,\delta\ln N + 1.$$

From the condition $x \leqslant N^{\varepsilon}$ we obtain the inequalities $z \leqslant \sqrt[3]{x} \leqslant N^{\varepsilon/3}$ and $\frac{\ln N}{\ln z} \geqslant 3\varepsilon^{-1} > 6$. Consequently,

$$r < \frac{1}{3}\,\delta\ln N + \frac{1}{6}\,\delta\ln N = \frac{1}{2}\,\delta\ln N = \frac{1}{2}\,z^{3(\beta-0.5)}\frac{\ln N}{\ln z}.$$

Returning to the estimation of V_2 we find that

$$V_2 < 2^{\nu}\frac{\ln N}{\ln z}\sum_{\substack{T-\sqrt{x}<\gamma\leqslant T+H+\sqrt{x}\\ \beta>0.5}}(\beta-0.5)^{\nu}(\xi^2 z^3)^{\beta-0.5}.$$

Using the fact that $\xi^2 z^3 \leqslant x$, as well as the identity

$$(\beta-0.5)^{\nu}x^{\beta-0.5} = \int_0^{\beta-0.5}(u^{\nu}\ln x + \nu u^{\nu-1})x^u\,du,$$

we transform the sum over γ as follows:

$$V_2 < 2^{\nu}\,\frac{\ln N}{\ln z}\sum_{\substack{T-\sqrt{x}<\gamma\leqslant T+H+\sqrt{x}\\ \beta>0.5}}\int_0^{\beta-0.5}(u^{\nu}\ln x + \nu u^{\nu-1})x^u\,du$$
$$= 2^{\nu}\,\frac{\ln N}{\ln z}\sum_{\substack{T-\sqrt{x}<\gamma\leqslant T+H+\sqrt{x}\\ \beta>0.5}}\int_0^{0.5} g(\beta+i\gamma;u)(u^{\nu}\ln x + \nu u^{\nu-1})x^u\,du,$$

where the function $g(\beta+i\gamma;u)$ is equal to 1 for $0 \leqslant u \leqslant \beta - 0.5$ and to 0 otherwise. We interchange the order of summation and integration in the last expression. Then the sum of the values $g(\beta+i\gamma;u)$ taken over all the ordinates γ in the interval $(T-\sqrt{x},\, T+H+\sqrt{x}\,]$ is equal to the difference

$$N\left(\frac{1}{2}+u, T+H+\sqrt{x}\right) - N\left(\frac{1}{2}+u, T-\sqrt{x}\right),$$

which, in turn, does not exceed the sum

$$\left(N\left(\frac{1}{2}+u, T+H\right) - N\left(\frac{1}{2}+u, T\right)\right) + \bigl(N(T+H+\sqrt{x}) - N(T+H)\bigr)$$
$$+ \bigl(N(T) - N(T-\sqrt{x})\bigr).$$

It follows from the density theorems stated above and the Riemann–von Mangoldt formula for $N(T)$ that the first summand is bounded by the quantity

$$13H(\ln T)x^{-2u} < 13 \cdot 2\pi(1+o(1))Mx^{-2u} < 81.7Mx^{-2u},$$

while the sum of the second and third summands is bounded by the quantity

$$\pi^{-1}\sqrt{x}(\ln T + O(1)) < 0.3Mx^{-2u}.$$

Thus, we obtain the estimate

$$\begin{aligned} V_2 &< 2^\nu \frac{\ln N}{\ln z} 82M \int_0^{0.5} (u^\nu \ln x + \nu u^{\nu-1}) x^u x^{-2u}\, du \\ &= 2^\nu \frac{\ln N}{\ln z} \frac{82M}{(\ln x)^\nu} \int_0^{0.5\ln x} (u^\nu + \nu u^{\nu-1}) e^{-u}\, du < 164\nu!\, 2^\nu \frac{\ln N}{\ln z} \frac{M}{(\ln x)^\nu}. \end{aligned}$$

Returning to the estimation of the original sum, we find that

$$\begin{aligned} V = V_1 + V_2 &\leqslant \frac{(2e)^{8m}M}{(\ln z)^\nu} + 164\nu!\, 2^\nu \frac{M}{(\ln x)^\nu} \frac{\ln N}{\ln z} \\ &< M\frac{(2e)^{8m}}{(\ln z)^\nu}\left(1 + \frac{\nu!}{18}\left(\frac{\ln z}{\ln x}\right)^\nu \frac{\ln N}{\ln z}\right). \end{aligned}$$

The lemma is proved.

Lemma 10. *Suppose that m is an integer, $1 \leqslant m < \frac{1}{192}\ln x$, and $x^{1/(4m)} < y \leqslant x^{1/m}$. Then we have the estimate*

$$R_{2m} = \sum_{N<n\leqslant N+M} \left(S(t_n) + \frac{1}{\pi}\sum_{p<y} \frac{\sin(t_n \ln p)}{\sqrt{p}}\right)^{2m} < \frac{1}{20}(Am^2)^m M,$$

where $A = e^{34}\varepsilon^{-3}$.

Proof. First, setting $z = x^{1/(12m)}$ we verify that

$$z > \exp\left(\frac{\ln x}{12}\frac{192}{\ln x}\right) = e^{16}.$$

As shown in § 2 of Ch. II in [7], for such z and an arbitrary t with the condition $t^2 > z$, the following inequality holds:

$$\left|\pi S(t) + \sum_{p<y} \frac{\sin(t\ln p)}{\sqrt{p}}\right| \leqslant \sum_{j=1}^{7} r_j, \tag{11}$$

where

$$r_1 = 48(\sigma_{z,t} - 0.5)\ln N, \qquad r_2 = \left|\sum_{p<z^3} \frac{\Lambda_z(p)}{\ln p}(1 - p^{0.5-\sigma_{z,t}})\frac{p^{it}}{\sqrt{p}}\right|,$$

$$r_3 = 15\pi(\sigma_{z,t} - 0.5)\Big|\sum_{p<z^3}\Lambda_z(p)(1-p^{0.5-\sigma_{z,t}})\,\frac{p^{it}}{\sqrt{p}}\Big|,$$

$$r_4 = 15\pi(\sigma_{z,t} - 0.5)\Big|\sum_{p<z^{1.5}}\Lambda_z(p)\,\frac{p^{it}}{\sqrt{p}}\Big|, \qquad r_5 = \frac{1}{2}\Big|\sum_{p<z^{1.5}}\frac{\Lambda_z(p^2)}{\ln p}\,\frac{p^{2it}}{p}\Big|,$$

$$r_6 = \Big|\sum_{p<z^3}\Big(1-\frac{\Lambda_z(p)}{\ln p}\Big)\frac{p^{it}}{\sqrt{p}}\Big|, \qquad r_7 = \Big|\sum_{z^3\leqslant p<y}\frac{p^{it}}{\sqrt{p}}\Big|.$$

We set $t = t_n$ in (11) and take the $2m$th power of both parts of the resulting inequality. By summing over n and applying Hölder's inequality we obtain the estimate

$$\pi^{2m}R_{2m} \leqslant 7^{2m-1}\sum_{j=1}^{7}W_j, \qquad W_j = \sum_{N<n\leqslant N+M} r_j^{2m}(t_n).$$

We estimate each of the sums W_j separately.

First we estimate W_1. By setting $\nu = 2m$, $\xi = 1$ in Lemma 9 and observing that $z^3\xi^2 \leqslant \sqrt[4]{x} \leqslant x$, we obtain

$$W_1 < 48^{2m}(\ln N)^{2m}\sum_{N<n\leqslant N+M}(\sigma_z(n)-0.5)^{2m}$$
$$< 48^{2m}(\ln N)^{2m}\frac{(2e)^{8m}M}{(\ln z)^{2m}}\Big(1+\frac{(2m)!}{18}\Big(\frac{\ln z}{\ln x}\Big)^{2m}\frac{\ln N}{\ln z}\Big).$$

Using the estimates $(2m)! < 4\sqrt{m}(\frac{2m}{e})^{2m}$ and $x \geqslant t_N^{\varepsilon/20} > N^{\varepsilon/21}$, we conclude that the second term in the parentheses does not exceed

$$\frac{4\sqrt{m}}{18}\Big(\frac{2m}{e}\Big)^{2m}\frac{252m\varepsilon^{-1}}{(12m)^{2m}} = \frac{56m^{3/2}}{(6e)^{2m}}\varepsilon^{-1} < (4\varepsilon)^{-1}.$$

Thus,

$$W_1 < (48\cdot 16e^4)^{2m}\Big(\frac{\ln N}{\ln z}\Big)^{2m}M(1+(4\varepsilon)^{-1}) < \frac{1}{3}\varepsilon^{-1}M(3\cdot 2^8e^4\cdot 252m\varepsilon^{-1})^{2m}$$
$$< \frac{1}{3}\varepsilon^{-(2m+1)}(e^{16.18}m)^{2m}M.$$

We now estimate W_2 and W_3. Giving values 0 and 1 to the parameter k we write down each of these sums in the form

$$U_k = (15\pi)^{2mk}\sum_{N<n\leqslant N+M}(\sigma_z(n)-0.5)^{2mk}V_k^{2m}(t_n),$$

where

$$V_k(t) = \Big|\sum_{p<z^3}\Lambda_z(p)(\ln p)^k\,\frac{1-p^{0.5-\sigma_{z,t}}}{\sqrt{p}\ln p}\,p^{it}\Big|.$$

Using the results of [7], [17] we conclude that

$$U_k \leqslant (15\pi)^{2mk}\sqrt{U'}\,\sqrt{U''},$$

where

$$U' = \sum_{N<n\leqslant N+M} (\sigma_z(n)-0.5)^{4m(k+1)} z^{4m(\sigma_z(n)-0.5)},$$

$$U'' = (\ln z)^{4m(k+1)+1}\int_0^\infty z^{-u}\sum_{N<n\leqslant N+M}\Big|\sum_{p<z^3}\frac{a(p)}{\sqrt{p}}\,p^{it_n}\Big|^{4m} du,$$

$$a(p) = \frac{\Lambda_z(p)(\ln p)^k \ln(pz)}{(\ln z)^{k+2}p^u}.$$

By setting $\nu = 4m(k+1)$ and $\xi = z^{4m}$ in Lemma 9 and observing that $z^3\xi^2 = z^{8m+3} \leqslant z^{11m} \leqslant x^{11/12} < x$, we find that

$$\begin{aligned} U' &< \frac{(2e)^{8m}M}{(\ln z)^{4m(k+1)}}\left(1+\frac{(4m(k+1))!}{18}\,\frac{12m}{(12m)^{4m(k+1)}}\,\frac{\ln N}{\ln x}\right)\\ &< \frac{(2e)^{8m}M}{(\ln z)^{4m(k+1)}}(1+(30\varepsilon)^{-1}) < (10\varepsilon)^{-1}M\frac{(2e)^{8m}}{(\ln z)^{4m(k+1)}}. \end{aligned}$$

Next, we verify that the sum under the integral sign in the expression for U'' satisfies the conditions of Lemma 7. Indeed, by setting

$$y = z^3 = x^{1/(4m)}, \qquad c = \frac{\ln x}{2\ln M},$$

we obtain

$$0 < c < \frac{1}{2}, \qquad M > \exp(8e^2\varepsilon^{-1}) > \exp(e^2c^{-1}),$$

$$ce^{-2}\ln M = \frac{1}{2e^2}\ln x > \frac{1}{96}\ln x > 2m.$$

By observing that $|a(p)| < 4\cdot 3^{k+1}\frac{\ln p}{\ln y}$, we find that

$$U'' < (\ln z)^{4m(k+1)}\cdot 3.2(2^6\cdot 3^{2(k+1)}m)^{2m}M.$$

Thus,

$$\begin{aligned} U_k &\leqslant (15\pi)^{2mk}\left(\frac{3.2}{10}\varepsilon^{-1}(2^4e^4\cdot 2^6\cdot 3^{2(k+1)}m)^{2m}M^2\right)^{1/2}\\ &< \frac{M}{\sqrt{\varepsilon}}(2^{10}\cdot 3^{2(k+1)}(15\pi)^{2k})^m m^m, \end{aligned}$$

whence for $k=0$ and $k=1$, respectively, we obtain

$$W_2 < \frac{M}{\sqrt{\varepsilon}}(e^{13.2}m)^m, \qquad W_3 < \frac{M}{\sqrt{\varepsilon}}(e^{23.1}m)^m.$$

We now estimate W_4. By applying Cauchy's inequality to this sum we obtain

$$W_4 \leqslant (15\pi)^{2m}\sqrt{U'}\,\sqrt{U''},$$

where

$$U' = \sum_{N<n\leqslant N+M} (\sigma_z(n) - 0.5)^{4m}, \qquad U'' = \sum_{N<n\leqslant N+M} \left| \sum_{p<z^{1.5}} \Lambda_z(p)\,\frac{p^{it_n}}{\sqrt{p}} \right|^{4m}.$$

The hypotheses of Lemma 9 hold if we set $\xi = 1$, $z = x^{1/(12m)}$, $\nu = 4m$. Consequently,

$$U' < M\frac{(2e)^{8m}}{(\ln z)^{4m}}\left(1 + \frac{(4m)!}{18}\,\frac{1}{(12m)^{4m-1}}\,\frac{\ln N}{\ln z}\right) < (1 + (50\varepsilon)^{-1})M\frac{(2e)^{8m}}{(\ln z)^{4m}}$$
$$< (10\varepsilon)^{-1}\frac{(2e)^{8m}M}{(\ln z)^{4m}}.$$

By setting $a(p) = \Lambda_z(p)\ln^{-1} z$ and $y = z^{1.5}$ in Lemma 7 and determining c from the relation $z^{1.5} = M^{c/(2m)}$, we now obtain

$$|a(p)| \leqslant \frac{3}{2}\frac{\ln p}{\ln y}, \qquad c = \frac{\ln x}{4\ln M} \leqslant \frac{1}{4}, \qquad ce^{-2}\ln M = \frac{1}{4e^2}\ln x > \frac{1}{96}\ln x > 2m,$$

and so

$$U'' \leqslant (\ln z)^{4m}\cdot 3.2M\left(2\cdot\frac{9}{4}\cdot 2m\right)^{2m}.$$

Thus,

$$W_4 \leqslant (15\pi)^{2m}\left(\frac{3.2}{10}(2^4e^4\cdot 3^2m)^{2m}\varepsilon^{-1}M^2\right)^{1/2} < \frac{M}{\sqrt{\varepsilon}}(e^{16.7}m)^m.$$

We now estimate W_5. By setting

$$a(p) = \frac{\Lambda_z(p^2)}{\ln p} \leqslant 1, \qquad y = z^{1.5}, \qquad c = \frac{\ln x}{8\ln M} < \frac{1}{2}$$

in Lemma 8, we find that $ce^{-2}\ln M = \frac{1}{2e^2}\ln x > \frac{1}{192}\ln x > m$, so that

$$W_5 \leqslant 2^{-2m}(2m)^m M < m^m M.$$

We now estimate W_6 and W_7. In the first case we set

$$a(p) = 1 - \frac{\Lambda_z(p)}{\ln p}, \qquad y = z^3, \qquad c = \frac{\ln x}{4\ln M}.$$

After verifying the inequalities

$$|a(p)| \leqslant 3\frac{\ln p}{\ln y}, \qquad ce^{-2}\ln M = \frac{1}{4e^2}\ln x > \frac{1}{192}\ln x > m, \qquad 0 < c \leqslant \frac{1}{4},$$

we use Lemma 7 to obtain the estimate

$$W_6 \leqslant 3.2(2\cdot 3^2m)^m M < 3.2(e^{2.9}m)^m M.$$

Finally, in the second case we take $a(p) = 0$ for $p \leqslant z^3$, and $a(p) = 1 = \frac{\ln p}{\ln y}\frac{\ln y}{\ln p}$ for $z^3 < p < y$, and observe that

$$|a(p)| \leqslant 4\frac{\ln p}{\ln y},$$

whence for $c = \frac{m\ln y}{\ln M}$ we have

$$c \leqslant \frac{\ln x}{\ln M} \leqslant \frac{\ln N^{\varepsilon/10}}{\ln N^{\varepsilon}} = \frac{1}{10}, \qquad ce^{-2}\ln M = me^{-2}\ln y \geqslant \frac{1}{4e^2}\ln x > \frac{1}{192}\ln x > m,$$

$$W_7 < 3.2(2\cdot 4^2 m)^m M < 3.2(e^{3.5}m)^m M.$$

Taking the sum of the estimates obtained, we find that

$$\begin{aligned} R_{2m} &< \frac{1}{7}M(7\pi^{-1})^{2m}(e^{16.8}m)^{2m}\varepsilon^{-(2m+1)} \\ &\quad\times\left(\frac{1}{3}+\varepsilon^{2.5}(e^{-19.1}+e^{-9.2}+e^{-15.6}+e^{-32.3})+3.2\varepsilon^3(e^{-29.4}+e^{-28.8})\right) \\ &< \frac{1}{20}(e^{34}m^2)^m\varepsilon^{-(2m+1)}M \leqslant \frac{1}{20}(e^{34}\varepsilon^{-3}m^2)^m M. \end{aligned}$$

The lemma is proved.

Corollary 3. *Suppose that k is an integer and $1 \leqslant k \leqslant \frac{1}{100}\ln x$. Then we have the estimate*

$$R_k = \sum_{N<n\leqslant N+M}\left|S(t_n)+\frac{1}{\pi}\sum_{p<y}\frac{\sin(t_n\ln p)}{\sqrt{p}}\right|^k < \frac{1}{5}(Ak^2)^{k/2}M.$$

Proof. It is sufficient to consider the case of odd k, $k = 2m-1$. Then the hypothesis implies that $m = \frac{k+1}{2} < \frac{1}{192}\ln x$. By applying Hölder's inequality to the sum R_k along with the assertion of Lemma 10, we have

$$\begin{aligned} R_k = R_{2m-1} &\leqslant M^{1/(2m)}R_{2m}^{1-1/(2m)} \leqslant M^{1/(2m)}\left(\frac{1}{20}(Am^2)^m M\right)^{1-1/(2m)} \\ &\leqslant \frac{1}{\sqrt{20}}(Am^2)^{m-1/2}M < \frac{1}{5}(Ak^2)^{k/2}M. \end{aligned}$$

The corollary is proved.

§ 3. The moments of the quantities $\Delta(n)$ with integer exponent

For $a \geqslant 0$ we set

$$\varkappa(a) = \frac{2}{\sqrt{\pi}}\int_0^\infty (2v)^a e^{-v^2}\,dv = \frac{2^a}{\sqrt{\pi}}\,\Gamma\left(\frac{a+1}{2}\right).$$

In particular, for an integer $m \geqslant 0$ we have the equations

$$\varkappa(2m) = \frac{(2m)!}{m!}, \qquad \varkappa(2m+1) = \frac{2^{2m+1}m!}{\sqrt{\pi}}.$$

Theorem 1. *Let $L = \ln\ln N$, let m be an integer with the condition $1 \leqslant m \leqslant \sqrt{L}$, and let*

$$I_{2m} = \sum_{N<n\leqslant N+M} \Delta^{2m}(n) = \sum_{N<n\leqslant N+M} S^{2m}(t_n).$$

Then the following formula holds:

$$I_{2m} = \frac{\varkappa(2m)}{(2\pi)^{2m}} M L^m (1 + \theta B^m L^{-0.5}),$$

where $B = e^{21}\varepsilon^{-1.5}$. In particular, we have the equation

$$\sum_{N<n\leqslant N+M} \Delta^2(n) = \frac{1}{2\pi^2}\, M \ln\ln N \left(1 + \frac{\theta B}{\sqrt{\ln\ln N}}\right).$$

Proof. We set $y = x^{1/(3m)}$,

$$U(t) = \sum_{p<y} \frac{p^{it}}{\sqrt{p}}, \qquad V(t) = \sum_{p<y} \frac{\sin(t\ln p)}{\sqrt{p}} = \frac{1}{2\pi i}(U(t) - \overline{U}(t)),$$

$$R(t) = S(t) + V(t).$$

Let a and b be arbitrary real numbers. Using Lagrange's finite increment formula along with Hölder's inequality, we obtain

$$\begin{aligned}(a+b)^{2m} &= a^{2m} + \theta \cdot 2mb(|a| + |b|)^{2m-1} \\ &= a^{2m} + \theta_1 \cdot 2m|b| \cdot 2^{2m-2}(|a|^{2m-1} + |b|^{2m-1}) \\ &= a^{2m} + \theta_1 \cdot 2^{2m-1} m(|a|^{2m-1}|b| + b^{2m}).\end{aligned}$$

By setting $a = -V(t_n)$, $b = R(t_n)$ and summing these equations over n, we find that

$$\begin{aligned}I_{2m} &= \sum_{N<n\leqslant N+M} S^{2m}(t_n) = \sum_{N<n\leqslant N+M} (V(t_n) - R(t_n))^{2m} \\ &= V_{2m} + \theta_2 m \cdot 2^{2m-1}(W_m + R_{2m}),\end{aligned}$$

where

$$V_{2m} = \sum_{N<n\leqslant N+M} V^{2m}(t_n), \qquad R_{2m} = \sum_{N<n\leqslant N+M} R^{2m}(t_n),$$

$$W_m = \sum_{N<n\leqslant N+M} |V(t_n)|^{2m-1}|R(t_n)|.$$

By applying Hölder's inequality to the last sum we obtain

$$\begin{aligned}W_m &\leqslant \left(\sum_{N<n\leqslant N+M} |V(t_n)|^{(2m-1)\cdot 2m/(2m-1)}\right)^{1-1/(2m)} \left(\sum_{N<n\leqslant N+M} |R(t_n)|^{2m}\right)^{1/(2m)} \\ &= V_{2m}^{1-1/(2m)} R_{2m}^{1/(2m)},\end{aligned}$$

$$I_{2m} = V_{2m} + \theta m \cdot 2^{2m-1}(V_{2m}^{1-1/(2m)} R_{2m}^{1/(2m)} + R_{2m}).$$

We further transform the sum V_{2m}:

$$V^{2m}(t) = \frac{(-1)^m}{(2\pi)^{2m}}(U(t) - \overline{U}(t))^{2m}$$
$$= (-1)^m(2\pi)^{-2m}\sum_{k=0}^{2m}(-1)^k\binom{2m}{k}U^k(t)\overline{U}^{2m-k}(t),$$
$$V_{2m} = (-1)^m(2\pi)^{-2m}\sum_{k=0}^{2m}(-1)^k\binom{2m}{k}v_k.$$

In the last expression, v_k denotes the sum

$$\sum_{N<n\leqslant N+M} U^k(t_n)\overline{U}^{2m-k}(t_n)$$
$$= \sum_{N<n\leqslant N+M}\sum_{\substack{p_1,\dots,p_k<y\\ q_1,\dots,q_{2m-k}<y}}(p_1\cdots q_{2m-k})^{-1/2}\left(\frac{p_1\cdots p_k}{q_1\cdots q_{2m-k}}\right)^{it_n}.$$

Lemma 5 and 6 imply the following relations for the quantities v_k:

$$v_k = \theta_k y^{3m}\ln N = \theta_k x\ln N, \quad k\neq m, \qquad v_m = Mv + \theta_m x\ln N,$$
$$v = m!\left(\sum_{p\leqslant y}p^{-1}\right)^m + \theta m^2 m!\left(\sum_p p^{-2}\right)\left(\sum_{p\leqslant y}p^{-1}\right)^{m-2}.$$

Since, as N increases unboundedly, we have the equation

$$x = \left(\frac{2\pi N}{\ln N}\right)^{a\varepsilon}(1+o(1))$$

for some a with the condition $\frac{1}{20}\leqslant a\leqslant\frac{1}{10}$, we conclude from Lagrange's finite increment formula that

$$\sum_{p\leqslant y}p^{-1} = L - \ln(m\varepsilon^{-1}) - 4.1\theta_1 = L - \theta_2\cdot 1.6\ln(m\varepsilon^{-1}),$$
$$\left(\sum_{p\leqslant y}p^{-1}\right)^m = L^m + \theta_3\cdot 1.6m\ln(m\varepsilon^{-1})L^{m-1}$$

and, consequently,

$$v - m!\,L^m = \theta\cdot 1.6m!\,m\ln(m\varepsilon^{-1})L^{m-1} + \theta\frac{1}{2}m^2m!\,L^{m-2}$$
$$= \theta\cdot 1.7m!\,m\ln(m\varepsilon^{-1})L^{m-1},$$
$$V_{2m} = (2\pi)^{-2m}\binom{2m}{m}Mv + \theta(2\pi)^{-2m}\sum_{k=0}^{2m}\binom{2m}{k}x\ln N$$
$$= \frac{(2m)!}{m!}\frac{ML^m}{(2\pi)^{2m}}(1+\theta\cdot 1.8m\ln(m\varepsilon^{-1})L^{-1}).$$

We now show that $\ln(m\varepsilon^{-1}) < \varepsilon^{-1}\ln(m+2)$ for all $m \geqslant 1$ and $0 < \varepsilon < \frac{1}{3}$. Indeed, for $m = 1$ this follows from the chain of obvious inequalities

$$3^{1/\varepsilon} > e^{1/\varepsilon} > 1 + \varepsilon^{-1} > \varepsilon^{-1},$$

and for $m \geqslant 2$ from the stronger estimate $\ln(m\varepsilon^{-1}) < \varepsilon^{-1}\ln m$, which is equivalent to the inequality

$$\ln m > \frac{\ln(\varepsilon^{-1})}{\varepsilon^{-1} - 1} = \frac{\varepsilon \ln \frac{1}{\varepsilon}}{1 - \varepsilon},$$

which in turn holds for all $m \geqslant 2$ and $0 < \varepsilon < \frac{1}{3}$. By observing that $m\ln(m+2) < 1.1m^{1.5}$, we finally find that

$$V_{2m} = \frac{\varkappa(2m)}{(2\pi)^{2m}} ML^m (1 + 2\theta\varepsilon^{-1} m^{1.5} L^{-1}).$$

Stirling's formula and the last relation for $N \geqslant N_0(\varepsilon)$ imply the estimate

$$V_{2m} < 2\,\frac{\varkappa(2m)}{(2\pi)^{2m}} ML^m < 3\left(\frac{m}{\pi^2 e}\right)^m ML^m.$$

Together with the estimate for the quantity R_{2m} given by Lemma 10, this gives rise to the following inequalities:

$$V_{2m}^{1-1/(2m)} R_{2m}^{1/(2m)} < 3\left(\frac{m}{\pi^2 e}\right)^{m-0.5} \sqrt{Am^2} ML^{m-0.5} < 3\pi\sqrt{Aem}\left(\frac{m}{\pi^2 e}\right)^m ML^{m-0.5},$$

$$\begin{aligned} 2^{2m-1} m V_{2m}^{1-1/(2m)} R_{2m}^{1/(2m)} &< \frac{3\pi\sqrt{e}}{2} m^{1.5}\sqrt{A}\left(\frac{4m}{\pi^2 e}\right)^m ML^{m-0.5} \\ &\leqslant \frac{3\pi\sqrt{e}}{2}\,\frac{m^{1.5}}{(\sqrt{e})^m}\left(\frac{4m\sqrt{A}}{\pi^2\sqrt{e}}\right)^m ML^{m-0.5} < 9.01\left(\frac{4m}{\pi^2}\sqrt{\frac{A}{e}}\right)^m ML^{m-0.5}. \end{aligned}$$

Returning to the expression for the sum I_{2m} we obtain

$$\begin{aligned} I_{2m} &= \frac{\varkappa(2m)}{(2\pi)^{2m}} ML^m (1 + 2\theta_1\varepsilon^{-1} m^{1.5} L^{-1}) + 9.01\theta_2\left(\frac{4m}{\pi^2}\sqrt{\frac{A}{e}}\right)^m ML^{m-0.5} \\ &\quad + \theta_3\,\frac{m}{40}\,(4Am^2)^m M = \frac{\varkappa(2m)}{(2\pi)^{2m}} ML^m (1 + \theta r_m), \end{aligned}$$

where

$$r_m = \frac{2\varepsilon^{-1} m^{1.5}}{L} + \frac{(2\pi)^{2m}}{\varkappa(2m)}\left(\frac{9.01}{\sqrt{L}}\left(\frac{4m}{\pi^2}\sqrt{\frac{A}{e}}\right)^m + \frac{m}{40}\left(\frac{4Am^2}{L}\right)^m\right).$$

Using the easily verifiable inequality

$$\frac{m!}{(2m)!} < 0.8\left(\frac{e}{4m}\right)^m,$$

we find that

$$r_m < \frac{2\varepsilon^{-1}m^{1.5}}{L} + \frac{m!}{(2m)!}\left(\frac{9.01}{\sqrt{L}}\left(16m\sqrt{\frac{A}{e}}\right)^m + \frac{m}{40}\left(\frac{16\pi^2Am^2}{L}\right)^m\right)$$
$$< \frac{2\varepsilon^{-1}m^{1.5}}{L} + \frac{7.3}{\sqrt{L}}(4\sqrt{Ae})^m + \frac{m}{50}\left(\frac{4\pi^2eAm}{L}\right)^m$$
$$= \frac{(4\sqrt{Ae})^m}{\sqrt{L}}\left(\frac{2\varepsilon^{-1}}{\sqrt{L}}\,\frac{m^{1.5}}{(4\sqrt{Ae})^m} + 7.3 + \frac{\sqrt{L}}{50}\,m\left(\frac{\pi^2m\sqrt{Ae}}{L}\right)^m\right).$$

Since the largest values of the fractions $\frac{m^{1.5}}{(4\sqrt{Ae})^m}$ and $m\big(\frac{\pi^2m\sqrt{Ae}}{L}\big)^m$ on the segment $1 \leqslant m \leqslant \sqrt{L}$ are attained at $m = 1$, it follows that for $N \geqslant N_0(\varepsilon)$ the estimate for the quantity r_m takes the form

$$r_m < \frac{(4\sqrt{Ae})^m}{\sqrt{L}}\left(\frac{2\varepsilon^{-1}}{\sqrt{L}}\,\frac{1}{4\sqrt{Ae}} + 7.3 + \frac{\sqrt{L}}{50}\,\frac{\pi^2\sqrt{Ae}}{L}\right) < e^2\,\frac{(4\sqrt{Ae})^m}{\sqrt{L}} < \frac{B^m}{\sqrt{L}},$$

where $B = e^{21}\varepsilon^{-1.5} > 4e^2\sqrt{Ae}$. The theorem is proved.

Theorem 2. *Let $L = \ln\ln N$, $A = e^{34}\varepsilon^{-3}$, let m be an integer with the condition $1 \leqslant m \leqslant \sqrt{L}$, and let*

$$J_{2m-1} = \sum_{N<n\leqslant N+M} \Delta^{2m-1}(n) = \sum_{N<n\leqslant N+M} S^{2m-1}(t_n).$$

Then we have the estimate

$$|J_{2m-1}| \leqslant \frac{3.5}{\sqrt{A}}(Am)^m ML^{m-1}.$$

Proof. Using the notation introduced in the proof of Theorem 1 and setting $a = -V(t_n)$ and $b = R(t_n)$ in the identity

$$(a+b)^{2m-1} = a^{2m-1} + \theta m\cdot 2^{2m-2}(a^{2m-2}|b| + |b|^{2m-1}),$$

we find that

$$J_{2m-1} = -V_{2m-1} + \theta m\cdot 2^{2m-2}(U_m + R_{2m-1}),$$

where

$$V_{2m-1} = \sum_{N<n\leqslant N+M} V^{2m-1}(t_n), \qquad R_{2m-1} = \sum_{N<n\leqslant N+M} |R(t_n)|^{2m-1},$$
$$U_m = \sum_{N<n\leqslant N+M} |V(t_n)|^{2m-2}|R(t_n)|.$$

By applying Hölder's inequality to the sum U_m we obtain

$$U_m \leqslant \left(\sum_{N<n\leqslant N+M} |V(t_n)|^{(2m-2)m/(m-1)}\right)^{1-1/m}\left(\sum_{N<n\leqslant N+M} |R(t_n)|^m\right)^{1/m}$$
$$= V_{2m}^{1-1/m}R_m^{1/m},$$

whence

$$J_{2m-1} = -V_{2m-1} + \theta m \cdot 2^{2m-2}(V_{2m}^{1-1/m} R_m^{1/m} + R_{2m-1}).$$

By using Corollary 3 and the estimate

$$V_{2m} < 3\left(\frac{m}{\pi^2 e}\right)^m ML^m$$

established in the proof of Theorem 1, we obtain

$$V_{2m}^{1-1/m} R_m^{1/m} < 3\left(\frac{m}{\pi^2 e}\right)^{m-1} ML^{m-1}\sqrt{A}m = 3\pi^2 e\sqrt{A}\left(\frac{m}{\pi^2 e}\right)^m ML^{m-1},$$

$$2^{2m-2}m(V_{2m}^{1-1/m} R_m^{1/m} + R_{2m-1}) < \frac{3\pi^2 e}{4}\sqrt{A}m\left(\frac{4m}{\pi^2 e}\right)^m ML^{m-1} + \frac{(16Am^2)^m}{40\sqrt{A}} M$$

$$= (Am)^m ML^{m-1}\left(\frac{3\pi^2 e}{4}\sqrt{A}m\left(\frac{4}{\pi^2 eA}\right)^m + \frac{L}{40\sqrt{A}}\left(\frac{16m}{L}\right)^m\right).$$

Since the largest values of the fractions in the parentheses are attained at $m = 1$, we finally find that

$$2^{2m-2}m(V_{2m}^{1-1/m} R_m^{1/m} + R_{2m-1}) < (Am)^m ML^{m-1}\left(\frac{3}{\sqrt{A}} + \frac{2}{5\sqrt{A}}\right)$$

$$= \frac{3.4}{\sqrt{A}}(Am)^m ML^{m-1}.$$

In order to estimate the sum V_{2m-1}, we set $y = x^{1/(3m)}$ and use Lemma 6:

$$V_{2m-1} = \sum_{N<n\leqslant N+M}\left(\frac{1}{2\pi i}\left(\sum_{p\leqslant y}\frac{p^{it_n}}{\sqrt{p}} - \sum_{q\leqslant y}\frac{q^{-it_n}}{\sqrt{q}}\right)\right)^{2m-1}$$

$$= (2\pi i)^{1-2m}\sum_{N<n\leqslant N+M}\sum_{k=0}^{2m-1}(-1)^{k+1}\binom{2m-1}{k}$$

$$\times \sum_{\substack{p_1,\dots,p_k\leqslant y\\ q_1,\dots,q_{2m-k-1}\leqslant y}}(p_1\cdots q_{2m-k-1})^{-0.5}\left(\frac{p_1\cdots p_k}{q_1\cdots q_{2m-k-1}}\right)^{it_n}$$

$$= (2\pi)^{1-2m}\sum_{k=0}^{2m-1}(-1)^{k+1}\binom{2m-1}{k}\theta y^{\frac{3}{2}(2m-1)}\ln N = \theta\pi^{1-2m}y^{3m-\frac{3}{2}}\ln N$$

$$= \theta x\ln N = \theta\,\frac{0.1}{\sqrt{A}}(Am)^m ML^{m-1}.$$

By taking the sum of the estimates obtained, we arrive at the assertion of the theorem.

§ 4. The moments of the quantities $|\Delta(n)|$ with fractional exponent

Theorem 3. *Let* $L = \ln\ln N$, $B = e^{21}\varepsilon^{-1.5}$, *and let* a *be an arbitrary number with the condition*

$$0 < a \leqslant \frac{e\ln L}{10B\ln\ln L}.$$

Then the following asymptotic formula holds:

$$I_a = \sum_{N<n\leqslant N+M} |\Delta(n)|^a = \frac{\varkappa(a)}{(2\pi)^a} ML^{0.5a}(1+\theta c(B^{-1}\ln L)^{-\gamma}).$$

Furthermore, in the case $0 < a \leqslant 1$ *the quantities* c *and* γ *can be set to be equal to* 90 *and* $0.5a$, *respectively, and in the case* $a > 1$ *to* $2^{15.4}$ *and* $\frac{1}{2} + \{\frac{a-1}{2}\}$. *In particular, for* $a = 1$ *we obtain*

$$\sum_{N<n\leqslant N+M} |\Delta(n)| = \pi^{-3/2} M\sqrt{\ln\ln N}(1+90\theta(B^{-1}\ln L)^{-0.5}).$$

Proof. We set $\delta(n) = 2\pi|\Delta(n)|L^{-0.5}$. Let

$$j(a) = M^{-1} \sum_{N<n\leqslant N+M} \delta^a(n).$$

Then the formulae of Theorem 1 for an integer m with the condition $1 \leqslant m \leqslant L^{0.5}$ imply the relations

$$j(2m) = \varkappa(2m)(1+\theta B^m L^{-0.5}).$$

First consider the case $0 < a \leqslant 1$. By using the identity

$$|\delta(n)|^a = \varkappa_1^{-1} \int_0^\infty \frac{\sin^2(\delta(n)u)}{u^{1+a}}\,du, \qquad \varkappa_1 = \varkappa_1(a) = \int_0^\infty \frac{\sin^2 u}{u^{1+a}}\,du,$$

and observing that

$$\varkappa_1 = \sum_{k=1}^\infty \int_{\pi(k-1)}^{\pi k} \frac{\sin^2 u}{u^{1+a}}\,du > \sum_{k=1}^\infty \frac{1}{(\pi k)^{1+a}} \int_0^\pi \sin^2 u\,du$$

$$= \frac{1}{2\pi^a} \sum_{k=1}^\infty \frac{1}{k^{1+a}} > \frac{1}{2\pi^a} \int_1^\infty \frac{du}{u^{1+a}} = \frac{1}{2\pi^a a} > \frac{1}{2\pi a},$$

we represent the sum $j(a)$ in the form

$$(M\varkappa_1)^{-1} \int_0^\infty \frac{1}{u^{1+a}} \sum_{N<n\leqslant N+M} \sin^2(\delta(n)u)\,du.$$

We set $\lambda = \frac{1}{4}(B^{-1}\ln L)^{0.5}$ and $K = [\frac{16}{3}e\lambda^2] + 1$ and approximate each of the functions $\sin^2(\delta(n)u)$ on the segment $0 \leqslant u \leqslant \lambda$ by the sum of the first K terms of the Taylor series. Since such expansions contain only even powers of $\delta(n)u$, after changing the order of summation we express the 'fractional moment' $j(a)$ as the

sum of the quantities $j(2k)$, $k = 0, 1, 2, \ldots, K$, taken with some coefficients. By further transforming the resulting sum using the formulae of Theorem 1, we obtain the desired formulae for $j(a)$ and I_a.

We now perform the necessary calculations. First, we have the equation $j(a) = j_1 + r_1$, in which

$$j_1 = \varkappa_1^{-1} \int_0^\lambda \frac{1}{u^{1+a}} \left(M^{-1} \sum_{N<n\leqslant N+M} \sin^2(\delta(n)u) \right) du,$$

$$r_1 = (M\varkappa_1)^{-1} \sum_{N<n\leqslant N+M} \int_\lambda^\infty \frac{\sin^2(\delta(n)u)}{u^{1+a}}\, du \leqslant \varkappa_1^{-1} \int_\lambda^\infty \frac{du}{u^{1+a}} < 2\pi\lambda^{-a}.$$

Next, the formula

$$\sin^2 x = \frac{1}{2} \sum_{k=1}^{K} \frac{(-1)^{k-1}}{(2k)!} (2x)^{2k} + \frac{\theta}{2} \frac{(2x)^{2K+2}}{(2K+2)!}$$

implies that $j_1 = j_2 + r_2$, where

$$j_2 = \varkappa_1^{-1} \int_0^\lambda \frac{1}{u^{1+a}} \frac{1}{2} \sum_{k=1}^{K} \frac{(-1)^{k-1}}{(2k)!}\, j(2k)\, du,$$

$$r_2 = \frac{1}{2}\theta\varkappa_1^{-1} \int_0^\lambda \frac{1}{u^{1+a}} \frac{(2u)^{2K+2}}{(2K+2)!}\, j(2K+2)\, du.$$

By observing that $B^{K+1} < B^2 L^{B^{-1}\ln B} < L^{0.1}$, it is easy to verify that the upper bounds for the quantities $j(2K+2)$ and $|r_2|$ do not exceed, respectively,

$$\varkappa(2K+2)(1+\theta L^{-0.4}) \leqslant 2\frac{(2K+2)!}{(K+1)!} \qquad \text{and}$$

$$\varkappa_1^{-1} \frac{4^{K+1}}{(K+1)!} \int_0^\lambda u^{2K+1-a}\, du \leqslant \left(\frac{4e\lambda^2}{K}\right)^K \lambda^{-a} \leqslant \left(\frac{3}{4}\right)^K \lambda^{-a}.$$

We transform the sum j_2 using Theorem 1. Denoting by r_3 the contribution to this sum of the remainder terms in the formulae for the quantities $j(2k)$, we obtain $j_2 = j_3 + r_3$, where

$$|r_3| < (2\varkappa_1)^{-1} B^K L^{-0.5} \int_0^\lambda \frac{1}{u^{1+a}} \sum_{k=1}^{K} \frac{(2u)^{2k}}{(2k)!} \frac{(2k)!}{k!}\, du < L^{-0.4}\lambda^{-a} \sum_{k=1}^{K} \frac{(2\lambda)^{2k}}{k!}$$
$$< L^{-0.4}\lambda^{-a} e^{4\lambda^2} < L^{-0.3}\lambda^{-a}.$$

Next, by replacing in the sum j_3 the quantities $\varkappa(2k)$ by their expressions in terms of improper integrals, we find that

$$j_3 = \varkappa_1^{-1} \int_0^\lambda \frac{1}{u^{1+a}} \left(\frac{1}{2} \sum_{k=1}^{K} \frac{(-1)^{k-1}}{(2k)!} (2u)^{2k} \int_0^\infty \frac{2}{\sqrt{\pi}} e^{-v^2} (2v)^{2k}\, dv \right) du$$
$$= \varkappa_1^{-1} \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-v^2} \int_0^\lambda \frac{1}{u^{1+a}} \left(\frac{1}{2} \sum_{k=1}^{K} \frac{(-1)^{k-1}}{(2k)!} (4uv)^{2k} \right) du\, dv.$$

We now observe that the sum over k in the last expression approximates the function $\sin^2(2uv)$ up to the summand $\theta\frac{(4uv)^{2K+2}}{(2K+2)!}$. Using this fact, we replace the integral over the segment $0 \leqslant u \leqslant \lambda$ by an improper integral. We obtain $j_3 = j_4 + r_4$, where

$$j_4 = \varkappa_1^{-1}\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-v^2}\int_0^\infty \frac{\sin^2(2uv)}{u^{1+a}}\,du\,dv = \varkappa_1^{-1}\varkappa_1\int_0^\infty e^{-v^2}(2v)^a\,dv = \varkappa(a),$$

$$\begin{aligned} r_4 &= \theta\varkappa_1^{-1}\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-v^2}\left(\int_0^\lambda \frac{(4uv)^{2K+2}\,du}{u^{1+a}(2K+2)!} + \int_\lambda^\infty \frac{\sin^2(2uv)}{u^{1+a}}\,du\right)dv \\ &= \theta\left(\varkappa_1^{-1}\frac{(2\lambda)^{2K+2}}{(2K+2)!}\varkappa(2K+2)\frac{\lambda^{-a}}{K} + 2\pi\lambda^{-a}\right) \\ &= \theta\left(\left(\frac{4e\lambda^2}{K}\right)^K\lambda^{-a} + 2\pi\lambda^{-a}\right) = \theta\lambda^{-a}\left(2\pi + \left(\frac{3}{4}\right)^K\right). \end{aligned}$$

By taking the sum of the estimates obtained for the quantities r_j, we finally find that

$$j(a) = \varkappa(a) + \theta\lambda^{-a}\left(4\pi + 2\left(\frac{3}{4}\right)^K + L^{-0.3}\right) = \varkappa(a)(1 + 90\theta(B^{-1}\ln L)^{-0.5a}).$$

We now consider the case $a > 1$. By representing a in the form $2m+1+b$, where $m \geqslant 0$ is an integer and $0 \leqslant b < 2$, we can use the identity

$$|\delta(n)|^{1+b} = \varkappa_2^{-1}\int_0^\infty \frac{\sin^4(\delta(n)u)}{u^{2+b}}\,du, \qquad \varkappa_2 = \varkappa_2(b) = \int_0^\infty \frac{\sin^4 u}{u^{2+b}}\,du > \frac{1}{5},$$

to write the original sum in the form

$$(M\varkappa_2)^{-1}\sum_{N<n\leqslant N+M}\delta^{2m}(n)\int_0^\infty \frac{\sin^4(\delta(n)u)}{u^{2+b}}\,du.$$

Following the same arguments as in the case $0 < a \leqslant 1$ and setting

$$\lambda = \frac{1}{16}\sqrt{B^{-1}\ln L}, \qquad K = [(12e\lambda)^2] + 1,$$

we find that $j(a) = j_1 + r_1$, where

$$j_1 = (M\varkappa_2)^{-1}\sum_{N<n\leqslant N+M}\delta^{2m}(n)\int_0^\lambda \frac{\sin^4(\delta(n)u)}{u^{2+b}}\,du,$$

$$r_1 = \theta\varkappa_2^{-1}\lambda^{-(1+b)}j(2m) = 5.1\theta\varkappa(a)\lambda^{-(1+b)}.$$

Next, let $b_k = \frac{(-1)^k}{2}(2^{2k-2}-1)$, $k \geqslant 2$, be the coefficients in the expansion of the function $\sin^4 x$ as a Taylor series in even powers of $2x$. Then we have the equation

$j_1 = j_2 + r_2$, where

$$r_2 = \frac{\theta}{2}\varkappa_2^{-1} j(2m+2K+2)\int_0^\lambda \frac{1}{u^{2+b}}\,\frac{(4u)^{2K+2}}{(2K+2)!}\,du$$
$$= 160\theta j(2m+2K+2)\left(\frac{\lambda}{K}\right)^2\left(\frac{16e^2\lambda^2}{4K^2}\right)^K \lambda^{-(1+b)},$$
$$j_2 = \varkappa_2^{-1}\int_0^\lambda \frac{1}{u^{2+b}}\left(\sum_{k=2}^K b_k \frac{(2u)^{2k}}{(2k)!}\,j(2m+2k)\right)du.$$

The easily verifiable inequalities $B^{m+K+1} < L^{0.1}$ and $2K^{m+1} < 2^K$ imply the estimates

$$j(2m+2K+2) < 2\varkappa(2m+2K+2) < 2(2(m+K+1))^{m+K+1}$$
$$< 2(3K)^{m+K+1} < 3^{2K}K^{m+1}K^K < (18K)^K,$$
$$|r_2| < \left(\frac{72e^2\lambda^2}{K}\right)^K < 2^{-K}.$$

We transform each of the sums $j(2m+2k)$ in the expression for j_2 using the formulae in Theorem 1. Denoting by r_3 the contribution of the remainder terms of these formulae, we find that $j_2 = j_3 + r_3$, where

$$j_3 = \varkappa_2^{-1}\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-v^2}(2v)^{2m}\int_0^\lambda \frac{1}{u^{2+b}}\left(\sum_{k=2}^K b_k\frac{(4uv)^{2k}}{(2k)!}\right)du\,dv,$$
$$r_3 = \theta B^m L^{-0.5}\int_0^\lambda \frac{1}{u^{2+b}}\left(\sum_{k=2}^K \frac{(16Bu^2)^k}{k!}\,\frac{\varkappa(2m+2k)}{\varkappa(2k)}\right)du.$$

To estimate r_3 we observe that for all $k \geqslant 2$,

$$\frac{\varkappa(2m+2k)}{\varkappa(2k)} = \binom{2m}{m}\binom{2m+2k}{2m}\binom{m+k}{m}^{-1} m! < 2^{2m}\cdot 2^{2m+2k}m! = 16^m m!\,2^{2k};$$

furthermore, $(16B)^m m! < L^{B^{-1}} < L^{0.1}$. Therefore the upper bound for r_3 does not exceed

$$L^{-0.4}\lambda^{-(1+b)}\sum_{k=2}^K \frac{(64B\lambda^2)^k}{k!} < L^{-0.4}e^{64B\lambda^2}\lambda^{-(1+b)} < L^{-0.15}\lambda^{-(1+b)}.$$

We now replace the sum over k by the quantity

$$\sin^4(2uv) + \frac{\theta}{2}\,\frac{(4uv)^{2K+2}}{(2K+2)!}$$

in the expression for j_3 and pass to an infinite interval of integration with respect to u. Thus we obtain $j_3 = j_4 + r_4$, where

$$j_4 = \varkappa_2^{-1}\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-v^2}(2v)^{2m}\int_0^\infty \frac{\sin^4(2uv)}{u^{2+b}}\,du\,dv$$
$$= \varkappa_2^{-1}\varkappa_2\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-v^2}(2v)^{2m+1+b}\,dv = \varkappa(a),$$

$$r_4 = \theta\varkappa_2^{-1}\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-v^2}\biggl((2v)^{2(m+K+1)}\int_0^\lambda \frac{(2u)^{2K+2}}{2u^{2+b}(2K+2)!}\,du + (2v)^{2m}\int_\lambda^\infty \frac{\sin^4(2uv)}{u^{2+b}}\,du\biggr)\,dv$$
$$= \theta\varkappa_2^{-1}\biggl(\frac{4}{\sqrt{K}}\Bigl(\frac{\lambda}{K}\Bigr)^2\Bigl(\frac{4e\lambda^2}{K}\Bigr)^K\frac{\varkappa(2m+2K+2)}{\varkappa(2K+2)}\,\lambda^{-(1+b)} + \varkappa(2m)\lambda^{-(1+b)}\biggr)$$
$$= \theta\varkappa_2^{-1}\biggl(\Bigl(\frac{16e\lambda^2}{K}\Bigr)^K\frac{16^m m!}{\sqrt{K}\varkappa(2m)} + 1\biggr)\varkappa(2m)\lambda^{-(1+b)}$$
$$= \theta\varkappa_2^{-1}\varkappa(2m)\biggl(4^m\Bigl(\frac{16e\lambda^2}{K}\Bigr)^K + 1\biggr)\lambda^{-(1+b)}$$
$$= \theta\varkappa_2^{-1}\varkappa(a)\biggl(1+\Bigl(\frac{32e\lambda^2}{K}\Bigr)^K\biggr)\lambda^{-(1+b)} = 5.1\,\theta\varkappa(a)\lambda^{-(1+b)}.$$

Thus, we arrive at the equations

$$j(a) = \varkappa(a) + \theta(2\cdot 5.1 + 2^{-K} + L^{-0.15})\varkappa(a)\lambda^{-(1+b)} = \varkappa(a)(1+10.3\,\theta\lambda^{-(1+b)})$$
$$= \varkappa(a)(1+2^{15.4}\theta(B^{-1}\ln L)^{-(1+b)/2}).$$

To complete the proof it remains to observe that for $a>1$,

$$\frac{1+b}{2} = \frac{1}{2} + \biggl\{\frac{a-1}{2}\biggr\} = \gamma.$$

The theorem is proved.

§ 5. Proof of Selberg's conjecture

Let $G(x)$ be the Gaussian distribution,

$$G(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^x e^{-u^2/2}\,du,$$

and let $g(t) = e^{-t^2/2}$ be the characteristic function of $G(x)$.

Theorem 4. *For real x let the quantity $\nu(x)$ denote the number of integers n, $N<n\leqslant N+M$, satisfying the condition*

$$\Delta(n) \leqslant x\sqrt{\frac{1}{2\pi^2}\ln\ln N}.$$

Then we have the equation

$$\nu(x) = M(G(x) + \theta C(\ln\ln\ln N)^{-0.5}), \qquad C = e^{22.4}\varepsilon^{-1.5}.$$

Proof. As above, we set $L = \ln\ln N$ and consider the characteristic function $f(t)$ of the discrete random quantity δ given by

$$\delta(n) = \pi\sqrt{\frac{2}{L}}\Delta(n), \qquad n = N+1,\dots,N+M,$$

that is, the sum

$$f(t) = M^{-1} \sum_{N<n\leqslant N+M} e^{it\delta(n)}.$$

Letting P be an integer with the condition $1 < P \leqslant \sqrt{L}$, we wish to approximate each of the exponentials $e^{it\delta(n)}$ by the sum of the first $2P$ terms of its Taylor series. We obtain

$$\begin{aligned} f(t) &= M^{-1} \sum_{N<n\leqslant N+M} \left(\sum_{k=0}^{2P-1} \frac{(it)^k}{k!}\, \delta^k(n) + 2\theta\, \frac{t^{2P}}{(2P)!}\, \delta^{2P}(n) \right) \\ &= \sum_{k=0}^{2P-1} \frac{(it)^k}{k!}\, j_k + 2\theta\, \frac{t^{2P}}{(2P)!}\, j_{2P}. \end{aligned}$$

Here j_k denotes the sum of the kth powers of the quantities $\delta(n)$ divided by M. It is easy to deduce from the formulae in Theorem 1 that $j_0 = 1$, and for $1 \leqslant k \leqslant P$,

$$j_{2k} = \frac{(2k)!}{k!}\, 2^{-k}(1 + \theta B^k L^{-0.5}) = \frac{(2k)!}{k!}\, 2^{-k}(1 + \theta_1 B^{2k} L^{-0.5}), \qquad B = e^{21}\varepsilon^{-1.5}.$$

Next, by setting $A = e^{34}\varepsilon^{-3} = e^{-8}B^2$ and using the estimate

$$\frac{k!}{(2k)!} < 0.8\left(\frac{e}{4k}\right)^k,$$

we find that

$$\begin{aligned} |j_{2k-1}| = (2\pi^2 L^{-1})^{k-0.5} M^{-1} |J_{2k-1}| &\leqslant \frac{7}{2\pi\sqrt{2}} (AL)^{-0.5} (2\pi^2 Ak)^k \\ &\leqslant \frac{7}{2\pi\sqrt{2}} (AL)^{-0.5} \frac{(2k)!}{k!}\, \frac{4}{5} \left(\frac{e}{4k}\right)^k (2\pi^2 Ak)^k \\ &< 0.5 L^{-0.5} \frac{(2k)!}{k!} \left(\frac{\pi^2}{2e^7} B^2\right)^k < 0.5 L^{-0.5} \frac{(2k)!}{k!}\, 2^{-k} B^{2k}. \end{aligned}$$

First we consider the summands in the expression for $f(t)$ corresponding to even k. We denote their contribution by $f_1(t)$. Then

$$\begin{aligned} f_1(t) &= \sum_{k=0}^{P-1} \frac{(it)^{2k}}{(2k)!} j_{2k} = \sum_{k=0}^{P-1} \frac{(-1)^k}{k!} \left(\frac{t^2}{2}\right)^k (1 + \theta B^{2k} L^{-0.5}) \\ &= g(t) + \frac{\theta}{P!} \left(\frac{t^2}{2}\right)^P + \theta L^{-0.5} \sum_{k=1}^{P-1} \frac{1}{k!} \left(\frac{B^2 t^2}{2}\right)^k. \end{aligned}$$

Next, let $f_2(t)$ be the contribution to $f(t)$ of the summands with k odd. Estimating the absolute value of $f_2(t)$, we obtain

$$|f_2(t)| \leqslant 0.5 L^{-0.5} \sum_{k=1}^{P} \frac{|t|^{2k-1}}{(2k-1)!}\, \frac{(2k)!}{k!} \left(\frac{B^2}{2}\right)^k = L^{-0.5} \sum_{k=0}^{P-1} \frac{|t|^{2k+1}}{k!} \left(\frac{B^2}{2}\right)^{k+1}.$$

Finally, we observe that

$$\frac{2t^{2P}}{(2P)!}j_{2P} \leqslant \frac{2t^{2P}}{P!}\cdot 2^{-P}(1+B^{2P}L^{-0.5}) \leqslant \frac{2}{P!}\left(\frac{t^2}{2}\right)^P + 2\cdot\frac{L^{-0.5}}{P!}\left(\frac{B^2t^2}{2}\right)^P.$$

Let $\lambda > 1$. Consider the integral

$$I(\lambda) = \int_0^\lambda |f(t)-g(t)|\frac{dt}{t}.$$

Using the estimates obtained above, we find that

$$\begin{aligned} I(\lambda) &\leqslant \int_0^\lambda \frac{3}{P!}\left(\frac{t^2}{2}\right)^P\frac{dt}{t} + L^{-0.5}\int_0^\lambda \sum_{k=1}^{P}\frac{1}{k!}\left(\frac{B^2t^2}{2}\right)^k\frac{dt}{t} \\ &\quad + L^{-0.5}\int_0^\lambda \sum_{k=0}^{P-1}\frac{1}{k!}\left(\frac{B^2}{2}\right)^{k+1}t^{2k}\,dt \\ &= \frac{1.5}{P\cdot P!}\left(\frac{\lambda^2}{2}\right)^P + L^{-0.5}\sum_{k=1}^{P}\frac{1}{2k\cdot k!}\left(\frac{B^2\lambda^2}{2}\right)^k \\ &\quad + \frac{1}{2}B^2\lambda L^{-0.5}\sum_{k=0}^{P-1}\frac{1}{k!(2k+1)}\left(\frac{B^2\lambda^2}{2}\right)^k \\ &< P^{-1.5}\left(\frac{e\lambda^2}{2P}\right)^P + B^2\lambda L^{-0.5}e^{0.5B^2\lambda^2}. \end{aligned}$$

We now set

$$\lambda = B^{-1}\sqrt{\frac{2}{3}\ln L}, \qquad P = [e\lambda^2]+1.$$

Then the integral $I(\lambda)$ does not exceed the quantity

$$P^{-1.5}2^{-P} + B^2\lambda L^{-1/6} < 2^{-P} + L^{-1/7} < L^{-\alpha}, \qquad \alpha = \frac{2}{3}B^{-2}e\ln 2 > e^{-42}\varepsilon^3.$$

By the Berry–Esseen inequality (see, for example, [23], Ch. VI, § 20.3), for any real x the modulus of the difference of the distribution functions $F(x)$ and $G(x)$ corresponding to the characteristic functions $f(t)$ and $g(t)$ does not exceed the sum

$$\frac{2}{\pi}I(\lambda) + \frac{24}{\pi\sqrt{2\pi}}\lambda^{-1} < \frac{2}{\pi}L^{-\alpha} + \frac{12\sqrt{3}}{\pi\sqrt{\pi}}\frac{B}{\sqrt{\ln L}} < \frac{C}{\sqrt{\ln L}},$$

where $C = e^{22.4}\varepsilon^{-1.5}$. Thus,

$$F(x) = G(x) + \frac{\theta C}{\sqrt{\ln L}}, \qquad \nu(x) = MF(x) = M\left(G(x) + \frac{\theta C}{\sqrt{\ln L}}\right).$$

Corollary 4. *Let $\Phi(n)$ be a monotonic positive function that increases unboundedly as $n \to +\infty$, and let m_1, m_2, m be equal, respectively, to the numbers of solutions of the inequalities*

$$|\Delta(n)| \leqslant \frac{1}{\Phi(n)}\sqrt{\ln\ln n}, \qquad |\Delta(n)| > \Phi(n)\sqrt{\ln\ln n},$$
$$\frac{1}{\Phi(n)}\sqrt{\ln\ln n} < |\Delta(n)| \leqslant \Phi(n)\sqrt{\ln\ln n},$$

where $N < n \leqslant N + M$. Then we have the relations

$$m_1 \leqslant 2M\biggl(\frac{2}{\sqrt{\pi}}\,\Phi^{-1} + q\biggr), \qquad m_2 < M\biggl(\frac{e^{-\pi^2\Phi^2}}{2\pi\Phi\sqrt{2}} + 2q\biggr),$$
$$m \geqslant M\biggl(1 - \frac{4}{\sqrt{\pi}}\,\Phi^{-1} - \frac{e^{-\pi^2\Phi^2}}{2\pi\Phi\sqrt{2}} - 4q\biggr),$$

where $\Phi = \Phi(N)$ and $q = q(N) = e^{22.4}\varepsilon^{-1.5}(\ln\ln\ln N)^{-0.5}$.

Proof. It follows from the monotonicity of $\Phi(n)$ that the number m_1 does not exceed the number of solutions of the inequality

$$|\Delta(n)| < \frac{2}{\Phi}\,\sqrt{\ln\ln N}.$$

Setting $x = 2\pi\sqrt{2}\Phi^{-1}$ we find that

$$m_1 \leqslant \nu(x) - \nu(-x) = M\bigl(G(x) - G(-x) + 2\theta q\bigr) = M\biggl(\frac{1}{\sqrt{2\pi}}\int_{-x}^{x} e^{-u^2/2}\,du + 2\theta q\biggr)$$
$$\leqslant M\biggl(x\sqrt{\frac{2}{\pi}} + 2q\biggr) = 2M\biggl(\frac{2\sqrt{\pi}}{\Phi} + q\biggr).$$

Similarly, m_2 does not exceed the number of solutions of the inequality

$$|\Delta(n)| > \Phi\sqrt{\ln\ln N} = y\sqrt{\frac{1}{2\pi^2}\ln\ln N},$$

where $y = \pi\Phi\sqrt{2}$. Consequently,

$$m_2 \leqslant \nu(+\infty) - \nu(y) = M\bigl(G(+\infty) - G(y) + 2\theta q\bigr) = M\biggl(\frac{1}{\sqrt{2\pi}}\int_{y}^{\infty} e^{-t^2/2}\,dt + 2\theta q\biggr)$$
$$\leqslant M\biggl(\frac{1}{\sqrt{2\pi}}\,\frac{e^{-y^2/2}}{y} + 2q\biggr) = M\biggl(\frac{e^{-\pi^2\Phi^2}}{2\pi\Phi\sqrt{2}} + 2q\biggr).$$

The last assertion of the corollary follows from the relation $m = M - m_1 - m_2$ and what was proved above. In particular, as $N \to +\infty$ we obtain $\Phi(N) \to +\infty$, $q(N) \to +\infty$, so that

$$m_1 \ll M\bigl(\Phi^{-1}(N) + q(N)\bigr) = o(M),$$
$$m_2 \ll M\bigl(\Phi^{-1}(N)e^{-\pi^2\Phi^2(N)} + q(N)\bigr) = o(M),$$
$$m = M - o(M).$$

Corollary 5. *The ordinate γ_n 'almost never' occurs in the Gram interval $(t_{n-1}, t_n]$, that is, Selberg's conjecture is true.*

Proof. First we show that the number of solutions of the double inequality

$$\gamma_n \leqslant t_n < \gamma_{n+1}, \quad \text{where} \quad N < n \leqslant N + M, \tag{12}$$

differs by at most 1 from the number of solutions of the double inequality

$$t_{n-1} < \gamma_n \leqslant t_n, \quad \text{where} \quad N < n \leqslant N + M. \tag{13}$$

Indeed, condition (12) is violated if and only if one of the two mutually exclusive inequalities $t_n < \gamma_n$, $\gamma_{n+1} \leqslant t_n$ holds, where $N < n \leqslant N + M$. Denoting the numbers of their solutions by k_1 and k_2, we find that the number of solutions of inequality (12) is $M - (k_1 + k_2)$.

Similarly, the number of solutions of inequality (13) is equal to $M - (n_1 + n_2)$, where n_1 and n_2 are the numbers of solutions, respectively, of the inequalities $t_n < \gamma_n$ and $\gamma_n \leqslant t_{n-1}$ with the condition $N < n \leqslant N + M$.

However, it is obvious that $k_1 = n_1$. The number n_2 is the number of solutions of the inequality $\gamma_{n+1} \leqslant t_n$, where $N - 1 < n \leqslant N + M - 1$, whence it differs from k_2 by at most 1. We showed above that the number of solutions of (12) is $o(M)$, whence the number of solutions of (13) is also $o(M)$.

Remark 1. For the problem of the distribution of the values of the function $S(t)$ as t varies over the segment $T < t \leqslant T + H$, the following analogue of Theorem 4 holds.

Theorem 5. *Let x be a real number and $E(x)$ the set of those $t \in (T, T + H]$ for which*

$$S(t) \leqslant x\sqrt{\frac{1}{2\pi^2}\ln\ln T}.$$

Then the following asymptotic formula holds:

$$\operatorname{mes} E(x) = H(G(x) + O((\ln\ln\ln T)^{-0.5})),$$

where the constant in the symbol O depends on ε.

This assertion is a strengthening of Theorem 6 in the survey [4], in which this asymptotics was obtained with remainder term of the order of $(\ln\ln\ln\ln T)^{-0.5}$. The proof of this assertion repeats practically word-for-word the proof of Theorem 4 and is based, among other things, on estimates for the integrals

$$\int_T^{T+H} S^k(t)\,dt, \qquad k = 1, 2, \ldots,$$

similar to those in Theorem 2.

As an application of Theorem 4 we obtain upper and lower bounds for the following sum, which is the 'moment of degree (-1)' of the quantities $|\Delta(n)|$:

$$I = I_{(-1)} = \sum_{N<n\leqslant N+M}{}' |\Delta(n)|^{-1}$$

(here and henceforth the prime on the sum symbol means summation over those n for which $\Delta(n)$ is non-zero).

Theorem 6. *We have the estimates*

$$\frac{M}{\sqrt{\ln\ln N}} \leqslant I \leqslant \frac{c\varepsilon^{-1.5}M}{\sqrt{\ln\ln\ln N}},$$

where $c > 0$ is an absolute constant.

Proof. We denote by k_ν the number of integers n in the sequence $N+1, \dots, N+M$ for which $|\Delta(n)| = \nu$, $\nu = 0, 1, 2, \dots$. We obtain

$$I = \frac{k_1}{1} + \frac{k_2}{2} + \frac{k_3}{3} + \cdots .$$

The sum on the right-hand side is finite since

$$|\Delta(n)| = |S(t_n)| \ll \ln t_n \ll \ln N,$$

and for any $r \geqslant 1$ it does not exceed the quantity

$$(k_1 + \cdots + k_r) + \frac{1}{r+1}(k_{r+1} + k_{r+2} + \cdots) \leqslant (k_1 + \cdots + k_r) + \frac{M}{r+1}.$$

The term $k_1 + \cdots + k_r$ is equal to the number of integers n for which $|\Delta(n)| \leqslant r$ and is of order not exceeding that of the quantity

$$M(r^{-1} + q(N)), \qquad q(N) = \frac{e^{21}\varepsilon^{-1.5}}{\sqrt{\ln\ln\ln N}}.$$

Setting $r = [\sqrt[4]{\ln\ln N}]$ we arrive at the inequality

$$I \ll M(r^{-1} + q(N)) + Mr^{-1} \ll \frac{\varepsilon^{-1.5}M}{\sqrt{\ln\ln\ln N}}.$$

In order to obtain a lower bound we sum both parts of the identity

$$\frac{1}{\delta(n)} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-\delta^2(n)x^2}\, dx$$

over n, where $\delta(n) = \pi\sqrt{\frac{2}{L}}\Delta(n)$. Replacing each of the exponentials by the quantity

$$1 - e^{-\theta(\delta(n)x)^2}(\delta(n)x)^2 = 1 - \theta_1(\delta(n)x)^2, \qquad 0 \leqslant \theta, \theta_1 \leqslant 1,$$

and the improper integral by the integral over the segment $0 \leqslant x \leqslant \alpha$, where $\alpha = \frac{1}{\sqrt{2}}$, we obtain

$$\begin{aligned} j &= \sum_{N<n\leqslant N+M}{}' |\delta(n)|^{-1} \geqslant \frac{2}{\sqrt{\pi}} \int_0^\alpha \sum_{N<n\leqslant N+M}{}' e^{-\delta^2(n)x^2}\,dx \\ &\geqslant \frac{2}{\sqrt{\pi}} \int_0^\alpha \sum_{N<n\leqslant N+M}{}' (1-\delta^2(n)x^2)\,dx \\ &= \frac{2}{\sqrt{\pi}} \int_0^\alpha \left(\sum_{N<n\leqslant N+M}{}' 1 - x^2 \sum_{N<n\leqslant N+M} \delta^2(n) \right) dx \\ &= \frac{2}{\sqrt{\pi}} \left(\alpha(M-k_0) - \frac{\alpha^3}{3} \frac{2\pi^2}{\ln\ln N} \frac{M}{2\pi^2} \ln\ln N \left(1 + \frac{B}{\sqrt{\ln\ln N}} \right) \right) \\ &\geqslant \frac{2}{\sqrt{\pi}} \left(\alpha(M-k_0) - \frac{\alpha^3}{3} M \left(1 + \frac{B}{\sqrt{\ln\ln N}} \right) \right). \end{aligned}$$

By Theorem 4 the quantity k_0 is of the order of $M(\ln\ln\ln N)^{-0.5}$. Therefore we finally obtain

$$\sum_{N<n\leqslant N+M}{}' |\delta(n)|^{-1} \geqslant 0.99 \cdot \frac{2}{\sqrt{\pi}} M \left(\alpha - \frac{\alpha^3}{3} \right) > \frac{1}{2} M,$$

$$I = \sum_{N<n\leqslant N+M}{}' |\Delta(n)|^{-1} = \frac{\pi\sqrt{2}}{\sqrt{\ln\ln N}} \sum_{N<n\leqslant N+M}{}' |\delta(n)|^{-1} > \frac{M}{\sqrt{\ln\ln N}}.$$

The theorem is proved.

Remark 2. The lower bound in Theorem 6 appears to be of the correct order, in that

$$\frac{c_1 M}{\sqrt{\ln\ln N}} \leqslant \sum_{N<n\leqslant N+M}{}' |\Delta(n)|^{-1} \leqslant \frac{c_2 M}{\sqrt{\ln\ln N}},$$

where c_1, c_2 are positive absolute constants.

Bibliography

[1] J.-P. Gram, "Note sur les zéros de la fonction $\zeta(s)$ de Riemann", *Acta Math.* **27**:1 (1903), 289–304.

[2] J. I. Hutchinson, "On the roots of the Riemann zeta function", *Trans. Amer. Math. Soc.* **27**:1 (1925), 49–60.

[3] E. C. Titchmarsh, "The zeros of the Riemann zeta function", *Proc. Roy. Soc. London Ser. A* **151** (1935), 234–255.

[4] E. C. Titchmarsh, "The zeros of the Riemann zeta function", *Proc. Roy. Soc. London Ser. A* **157** (1936), 261–263.

[5] A. Selberg, "The zeta function and the Riemann hypothesis", *C. R. Dixième Congrès Math. Skandinaves* 1946, vol. 10, Jul. Gjellerups Forlag, Copenhagen 1947, pp. 187–200.

[6] A. A. Karatsuba and M. A. Korolev, "The argument of the Riemann zeta function", *Uspekhi Mat. Nauk* **60**:3 (2005), 41–96; English transl., *Russian Math. Surveys* **60**:3 (2005), 433–488.

[7] A. A. Karatsuba and M. A. Korolev, "Behaviour of the argument of the Riemann zeta function on the critical line", *Uspekhi Mat. Nauk* **61**:3 (2006), 3–92; English transl., *Russian Math. Surveys* **61**:3 (2006), 389–482.

[8] A. Ghosh, "On the Riemann zeta function – mean value theorems and the distribution of $|S(t)|$", *J. Number Theory* **17**:1 (1983), 93–102.

[9] E. C. Titchmarsh, "The mean-value of the zeta function on the critical line", *Proc. London Math. Soc.* (2) **27**:1 (1928), 137–150.

[10] J. Moser, "On Gram's law in the theory of the Riemann zeta-function", *Acta Arith.* **32**:2 (1977), 107–113. (Russian)

[11] J. Moser, "A proof of E. C. Titchmarsh's conjecture in the theory of the Riemann zeta function", *Acta Arith.* **36**:2 (1980), 147–156. (Russian)

[12] J. Moser, "On a biquadratic sum in the theory of the Riemann zeta-function", *Acta Math. Univ. Comenian.* **42–43** (1983), 35–39. (Russian)

[13] A. Fujii, "A comparative study of the zeros of Dirichlet L-functions", *Bull. Amer. Math. Soc.* **81** (1975), 199–201.

[14] A. Fujii, "On the zeros of Dirichlet L-functions. V", *Acta Arith.* **28**:4 (1976), 395–403.

[15] A. Fujii, "Gram's law for the zeta zeros and the eigenvalues of Gaussian unitary ensembles", *Proc. Japan Acad. Ser. A Math. Sci.* **63**:10 (1987), 392–395.

[16] A. A. Lavrik, "Titchmarsh's problem in the discrete theory of the Riemann zeta function", *Number theory and analysis*, Trudy Mat. Inst. Steklov., vol. 207, Nauka, Moscow 1994, pp. 197–230; English transl., *Proc. Steklov Inst. Math.* **207**:6 (1995), 179–209.

[17] A. Selberg, "Contributions to the theory of the Riemann zeta function", *Arch. Math. Naturvid.* **48**:5 (1946), 89–155.

[18] A. Ghosh, "On Riemann's zeta function – sign changes of $S(T)$", *Recent progress in analytic number theory. Vol.* I (Durham 1979), Academic Press, London–New York 1981, pp. 25–46.

[19] A. A. Karatsuba, "On the function $S(t)$", *Izv. Ross. Akad. Nauk Ser. Mat.* **60**:5 (1996), 27–56; English transl., *Izv. Math.* **60**:5 (1996), 901–931.

[20] M. A. Korolev, "The argument of the Riemann zeta function on the critical line", *Izv. Ross. Akad. Nauk Ser. Mat.* **67**:2 (2003), 21–60; English transl., *Izv. Math.* **67**:2 (2003), 225–264.

[21] S. M. Voronin and A. A. Karatsuba, *The Riemann zeta function*, Fizmatlit, Moscow 1994; English transl., de Gruyter Exp. Math., vol. 5, de Gruyter, Berlin 1992.

[22] I. M. Vinogradov, *The method of trigonometric sums in number theory*, 2nd ed., Nauka, Moscow 1980; English transl. of 1st ed., *Trigonometrical sums in number theory*, Statistical Publ. Society, Calcutta 1978.

[23] M. Loève, *Probability theory*, Van Nostrand, Princeton–Toronto–New York–London 1960; Russian transl., Inostr. Lit., Moscow 1962.

M. A. Korolev
Steklov Mathematical Institute, RAS
E-mail: hardy_ramanujan@mail.ru

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